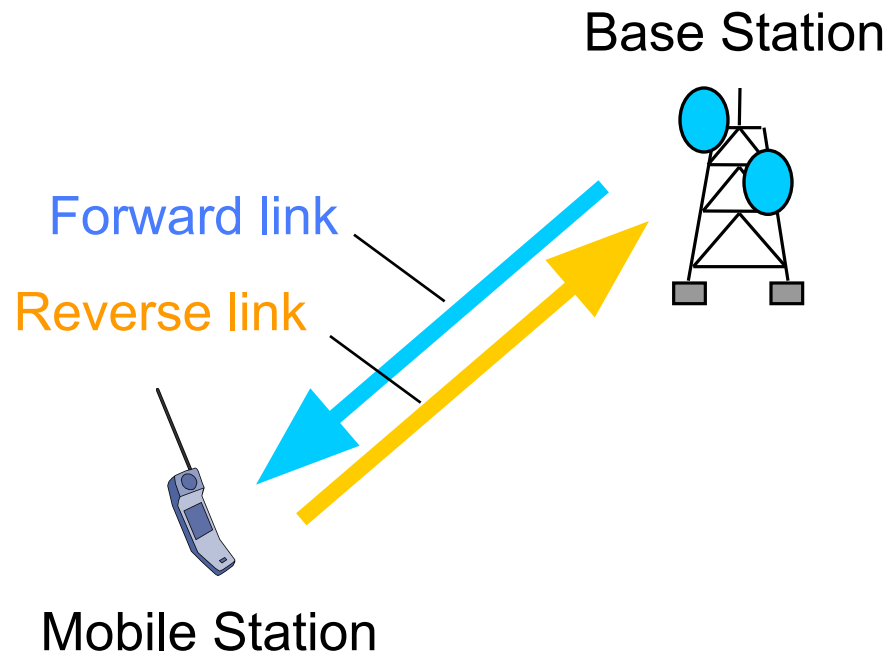


MULTIPLE ACCESS METHOD

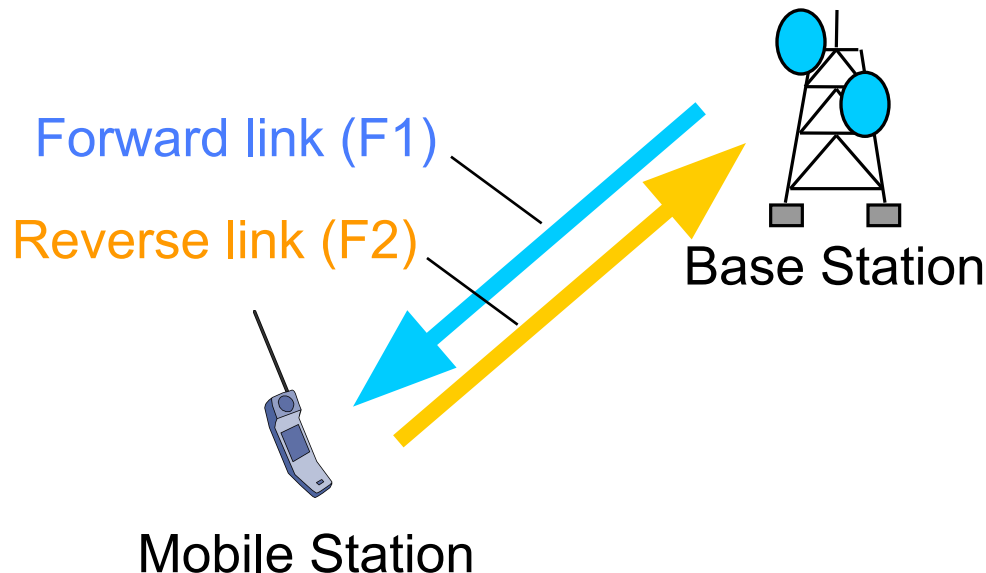
*Presented by
Dhiraj Patel*

Duplex Methods of Radio Links



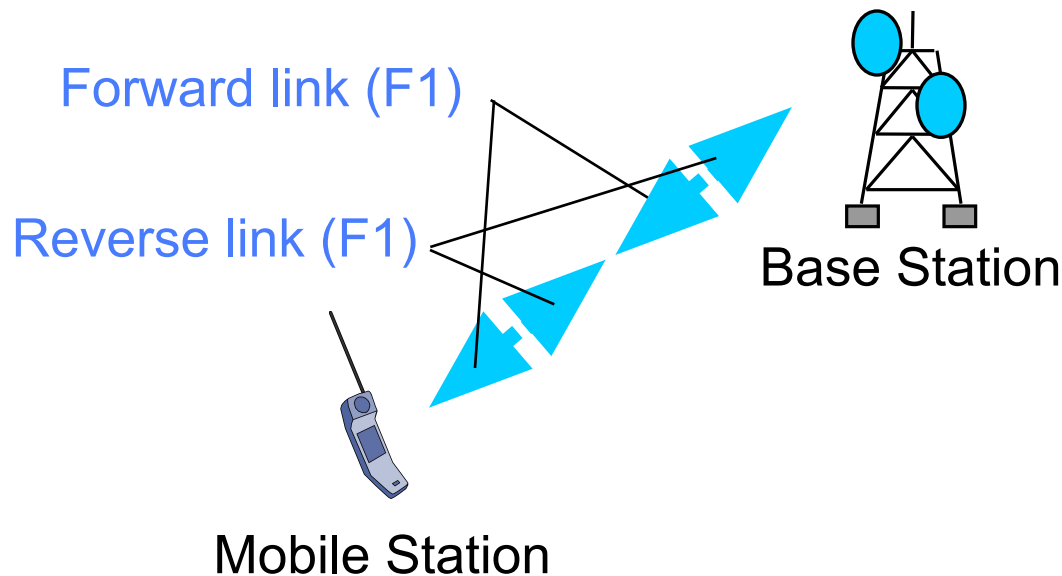
Frequency Division Duplex (FDD)

- Forward link frequency and reverse link frequency is different
- In each link, signals are continuously transmitted in parallel.

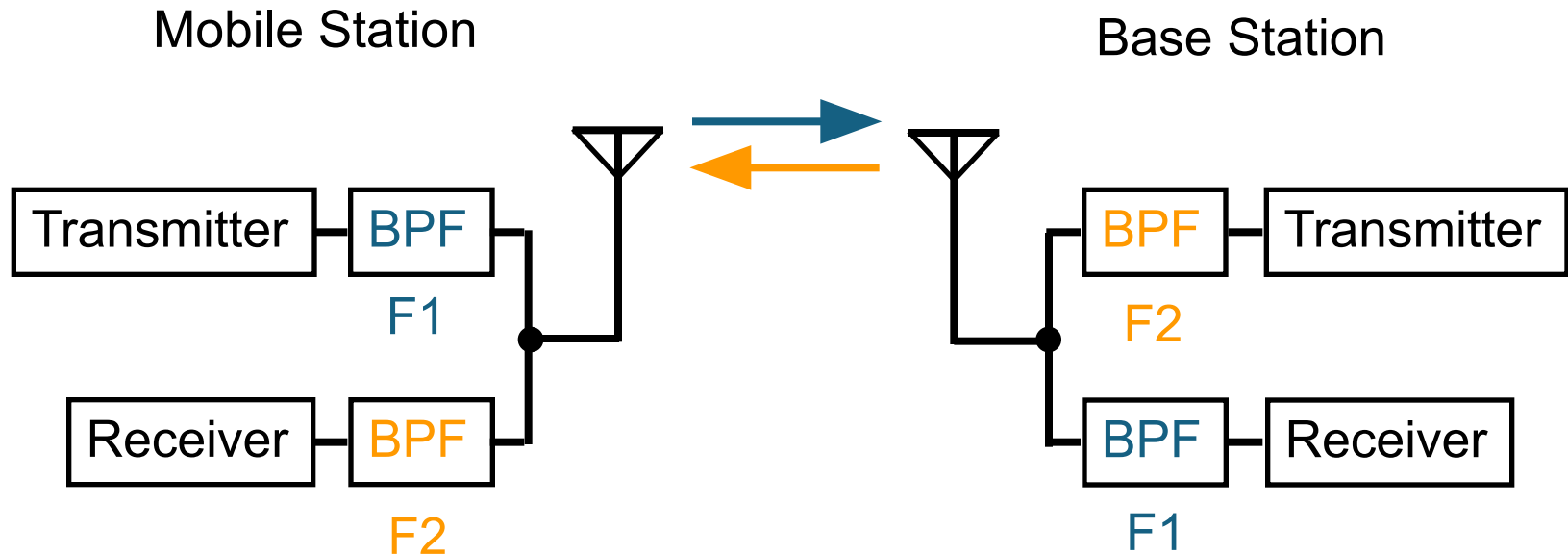


Time Division Duplex (TDD)

- Forward link frequency and reverse link frequency is the same.
- In each link, signals are incontinuously transmitted by turns just like a ping-pong.

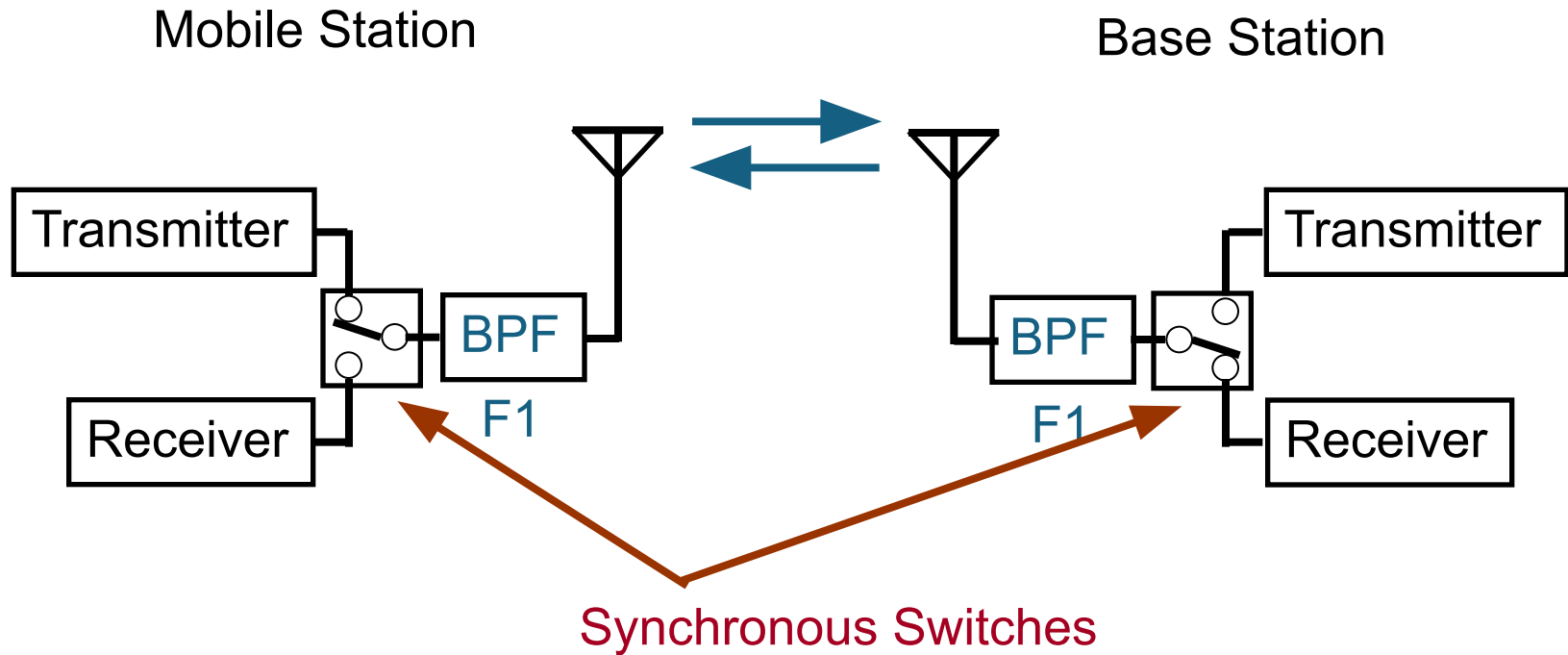


Example of FDD systems



BPF: Band Pass Filter

Example of TDD Systems



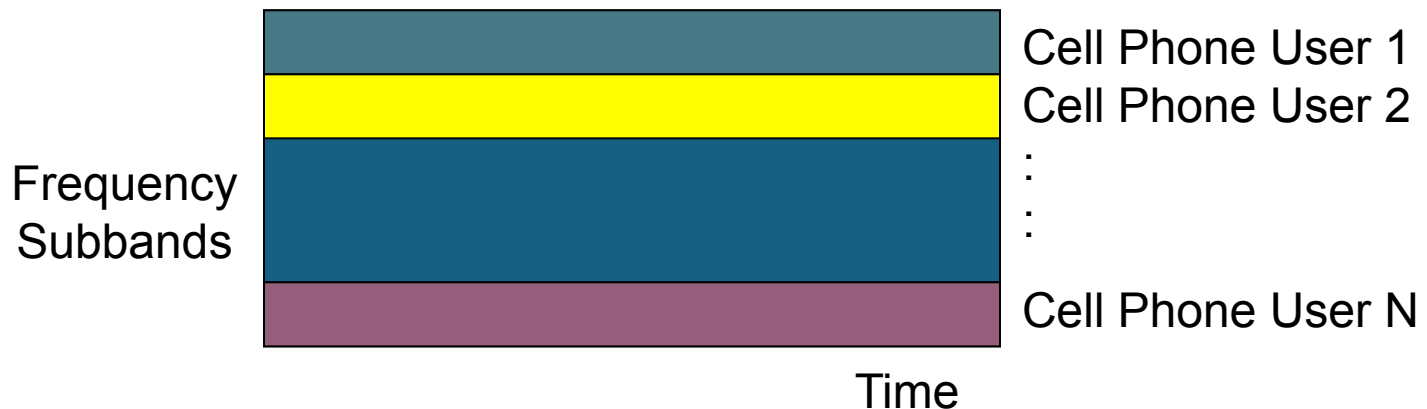
BPF: Band Pass Filter

Multiple Access Method

- In such cases, system adopts a multiple access policy.
- Three widely-used policies:
 - Frequency Division Multiple Access (FDMA)
 - Time Division Multiple Access (TDMA)
 - Code Division Multiple Access (CDMA)

FDMA

- In FDMA, we assume that a base station can receive radio signals in a given band of spectrum, i.e., a range of continuous frequency values.
- The band of frequency is broken up into smaller bands, i.e., subbands.
- Each transmitter (user) transmits to the base station using radio waves in its own subband.



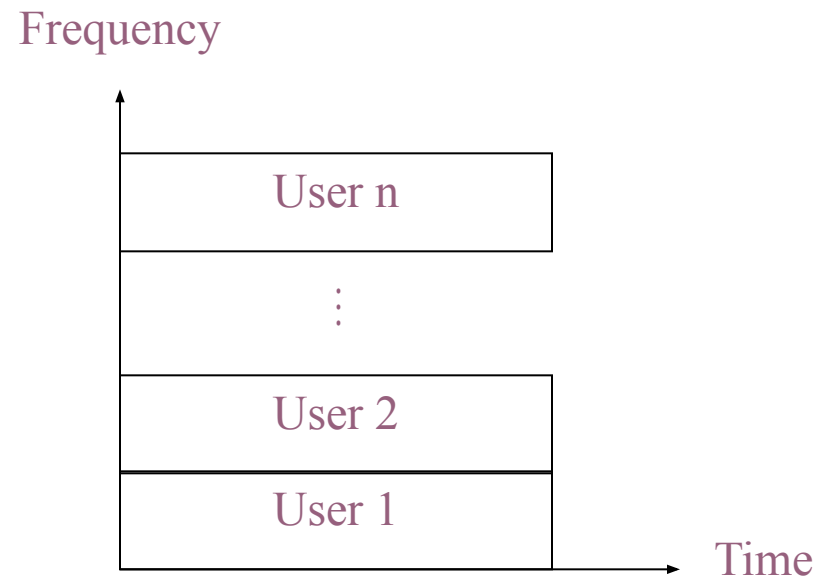
FDMA (Cont'd)

- A subband is also a range of continuous frequencies, e.g., 824 MHz to 824.1 MHz. The width of this subband is $0.1 \text{ MHz} = 100 \text{ KHz}$.
- When a user is assigned a subband, it transmits to the base station using a sine wave with the **center frequency** in that band, e.g., 824.05 MHz.

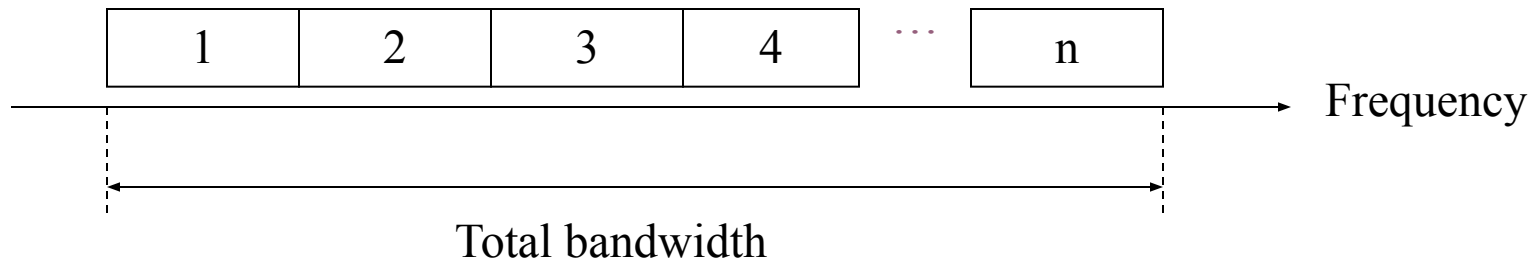
FDMA (Cont'd)

- When the base station is tuned to the frequency of a desired user, it receives no portion of the signal transmitted by another in-cell user (using a different frequency).
- This way, the multiple local transmitters within a cell do not interfere with each other.

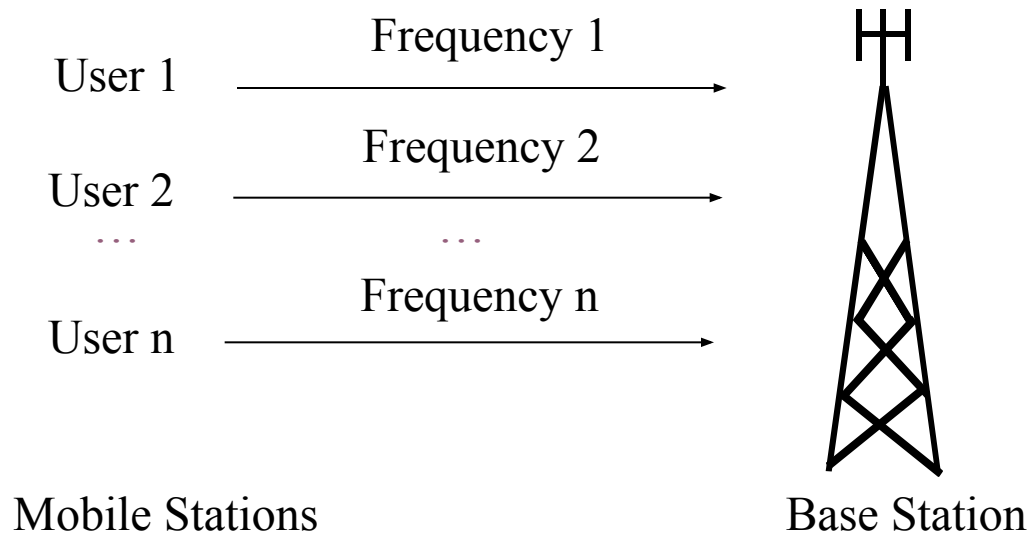
FDMA (Frequency Division Multiple Access)



FDMA Bandwidth Structure



FDMA Channel Allocation



- Lower Complexity
- Fewer bit for overhead purpose.
- High Cell site cost
- Tight RF filtering.

- Nonlinear Effect in FDMA

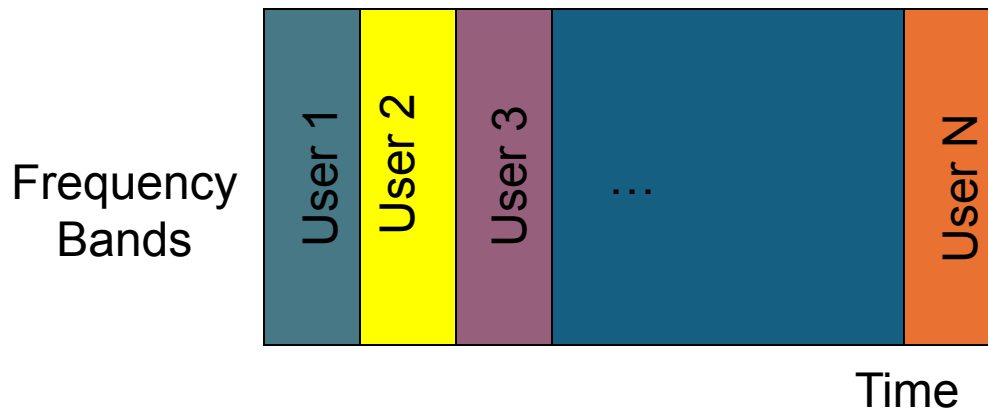
(B,coz many user share same antenna @ BS ,the power amplifier operated near saturation for maximum power efficiency causes nonlinearity.

Thus spreading in frequency domain and generate intermodulation frequency(IM).

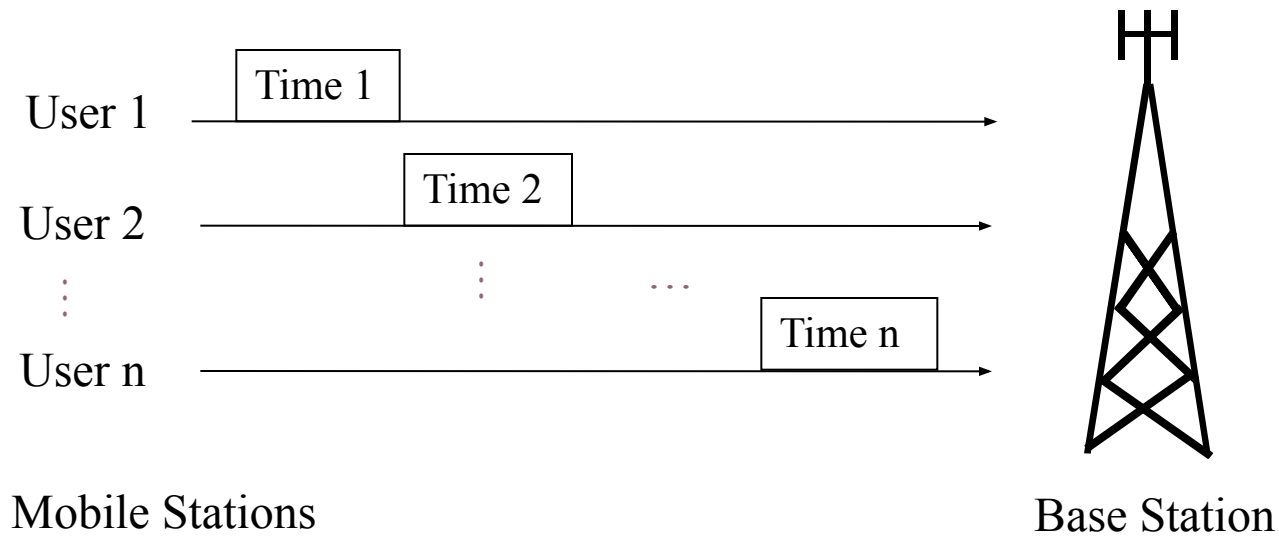
Produces undesired harmonics.

TDMA

- In pure TDMA, base station does not split up its allotted frequency band into smaller frequency subbands.
- Rather it communicates with the users one-at-a-time, i.e., “round robin” access.



TDMA Frame Illustration for Multiple Users

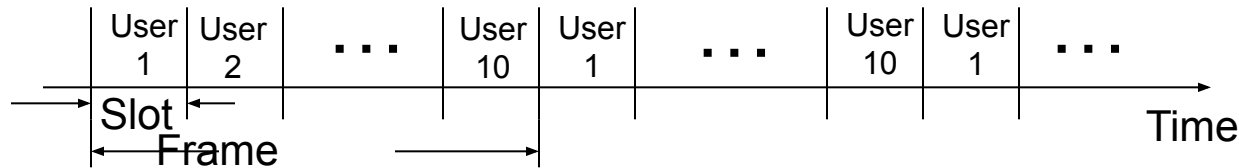


TDMA (Cont'd)

- Time is broken up into time slots, i.e., small, equal-length intervals.
- Assume there are some n users in the cell.
- Base station groups n consecutive slots into a frame.
- Each user is assigned one slot per frame. This slot assignment stays fixed as long as the user communicates with the base station (e.g., length of the phone conversation).

TDMA (Cont'd)

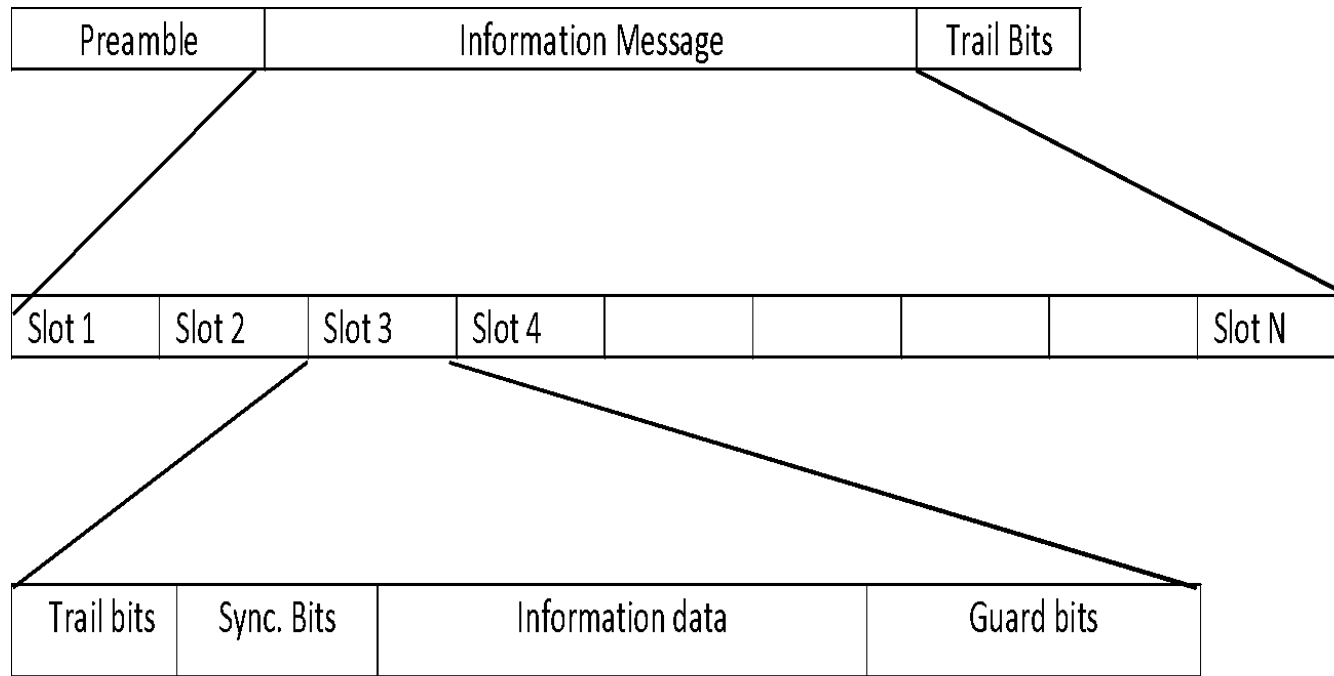
- Example of TDMA time slots for $n = 10$.



- In each time slot, the assigned user transmits a radio wave using a sine wave at the center frequency of the frequency band assigned to the base station.

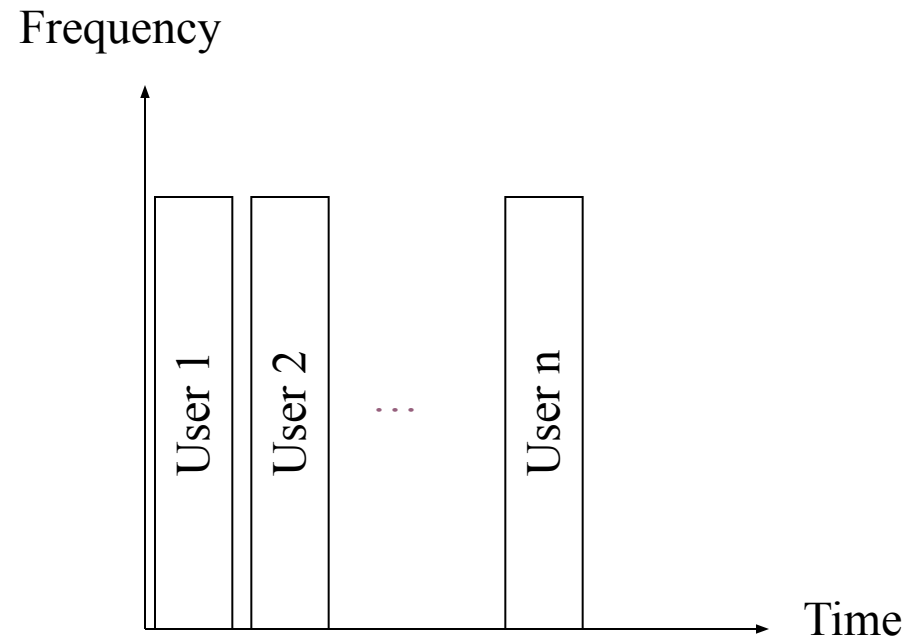
Channels

- Channel is a general term which refers to a frequency in an FDMA system, a timeslot/frequency combination in TDMA, or a code in CDMA.
- This way, a base station has a fixed number of channels and can support only that many simultaneous users.

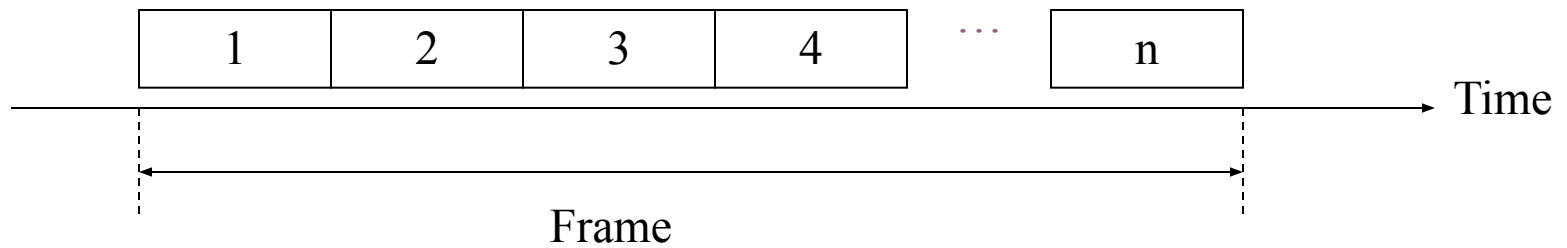


TDMA frame structure

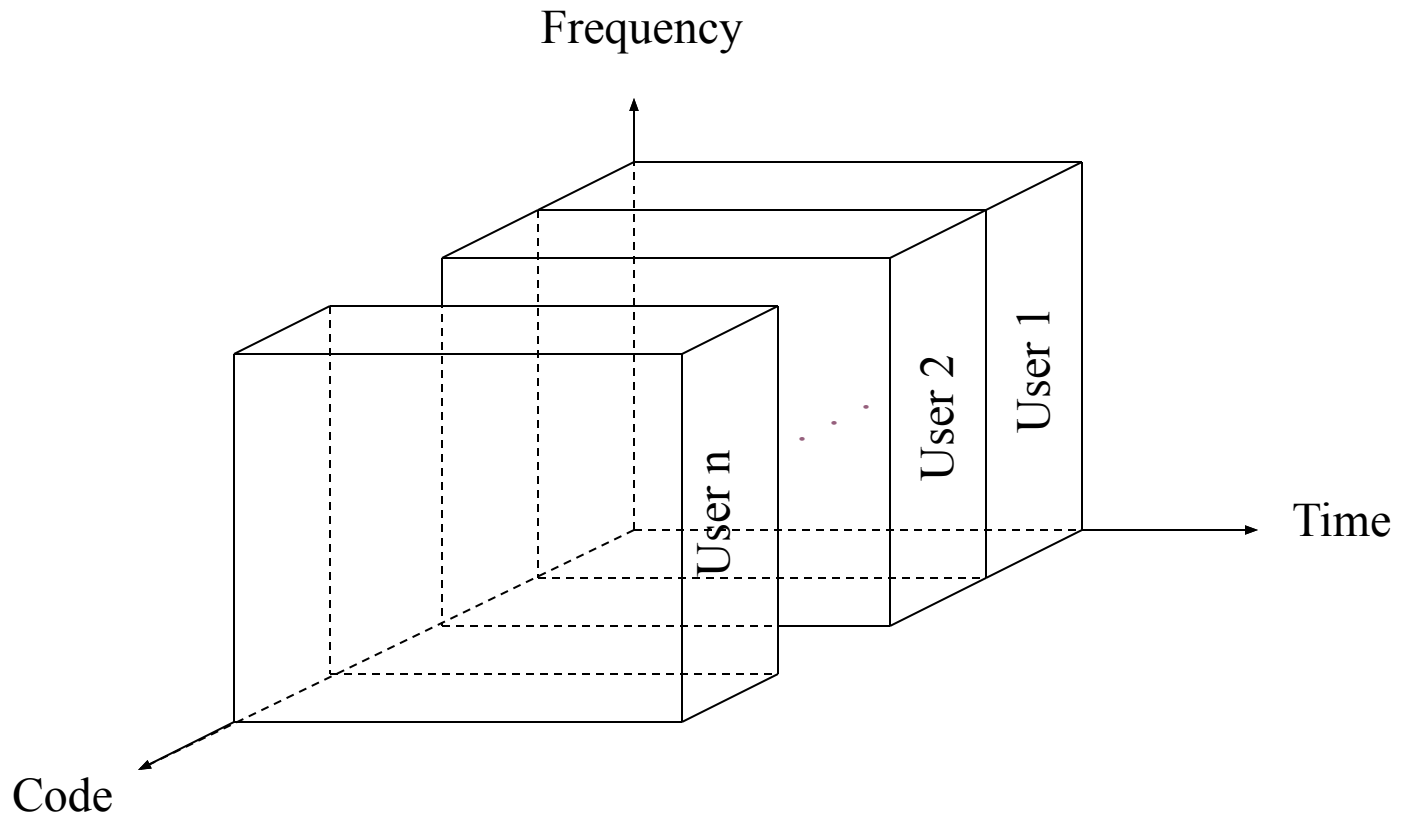
TDMA (Time Division Multiple Access)



TDMA Frame Structure



CDMA (Code Division Multiple Access)



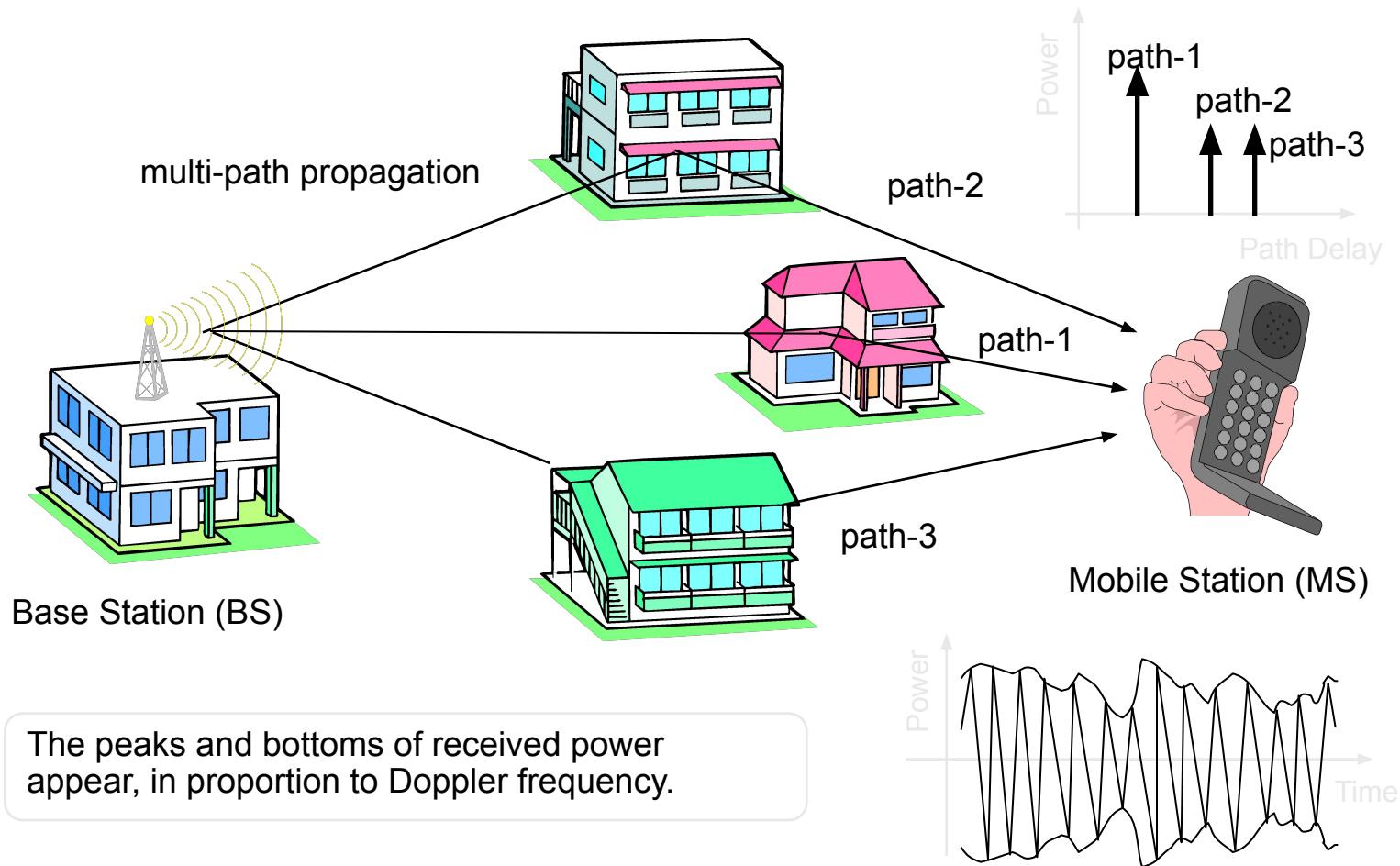
CDMA

- CDMA is a more complicated scheme.
- Narrowband msg. signal is multiplied by a very large B.W. signal, called spreading signal
- Here all users communicate to the receiver at the same time and using the same set of frequencies.
- This means they may interfere with each other.
- The system is designed to control this interference.
- A desired user's signal is deciphered using a unique code assigned to the user.

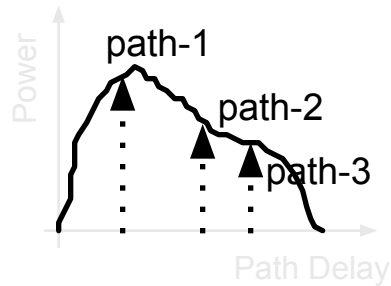
- @ receiver it performs a time correlation operation to detect only specific desired code word.
- Near far problem.
- Power control implementation.
- FDD/CDMA or TDD/CDMA could be used.
- Soft capacity limit(limited users).
- Multipath fading is reduced(SS B.W.> coherence B.W., inherent frequency diversity effect is neglected by small scale fading)

Features of CDMA

Mobile Propagation Environment : Multi-path Fading



Fading in non-CDMA System



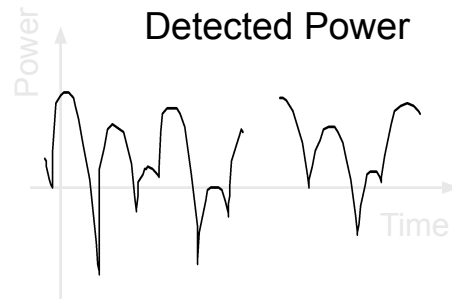
With low time-resolution,
different signal paths cannot be discriminated.

...

These signals sometimes strengthen,
and sometimes cancel out each other,
depending on their phase relation.

... This is “fading”.

...

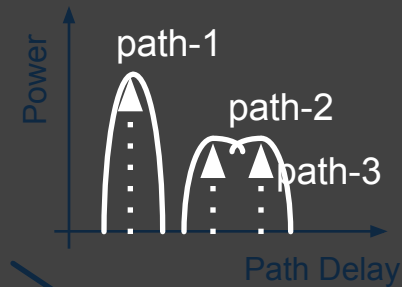


In this case, signal quality is damaged
when signals cancel out each other.
In other words, signal quality is dominated
by the probability for detected power
to be weaker than minimum required level.

This probability exists with less than two paths.

In non-CDMA system, “fading” damages signal quality.

Fading in CDMA System ...

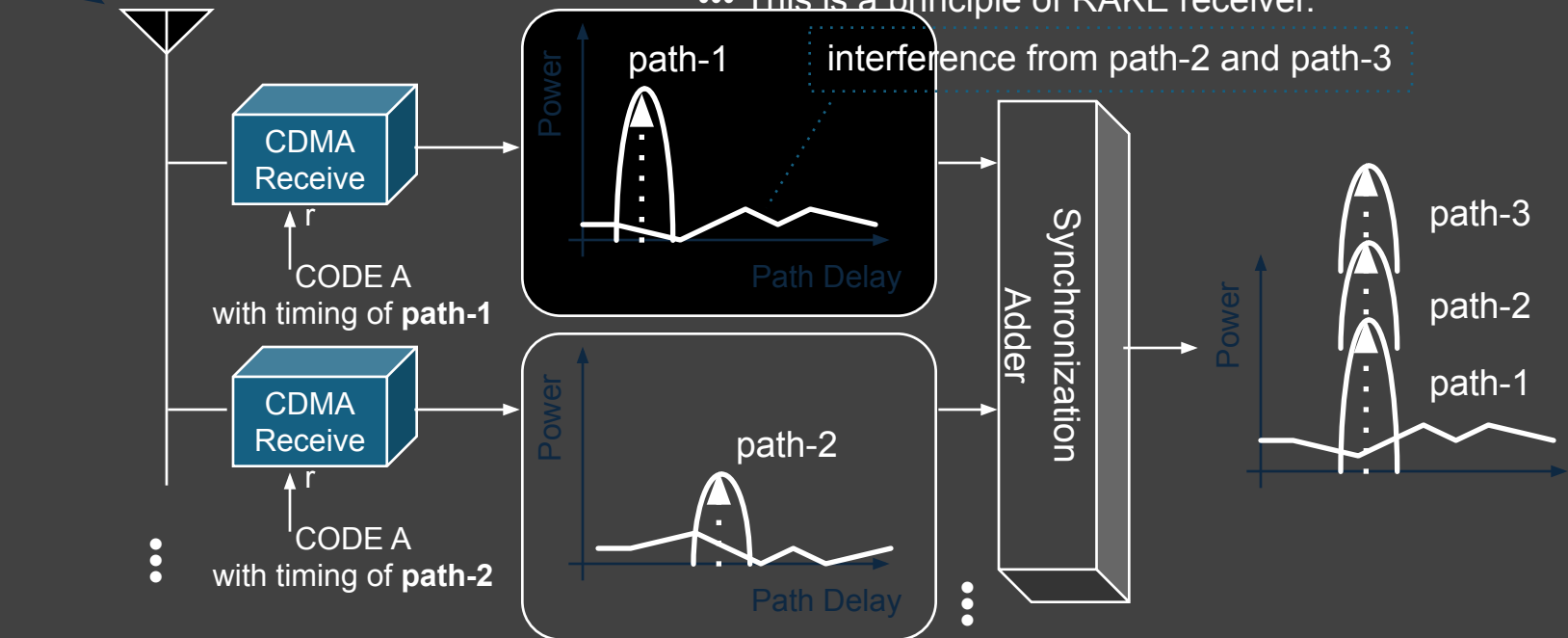


Because CDMA has high time-resolution,
different path delay of CDMA signals
can be discriminated.

...

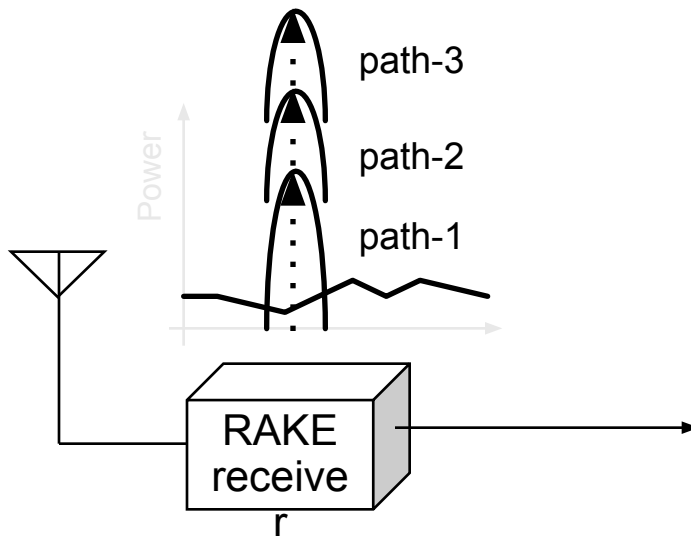
Therefore, energy from all paths can be summed
by adjusting their phases and path delays.

... This is a principle of RAKE receiver.



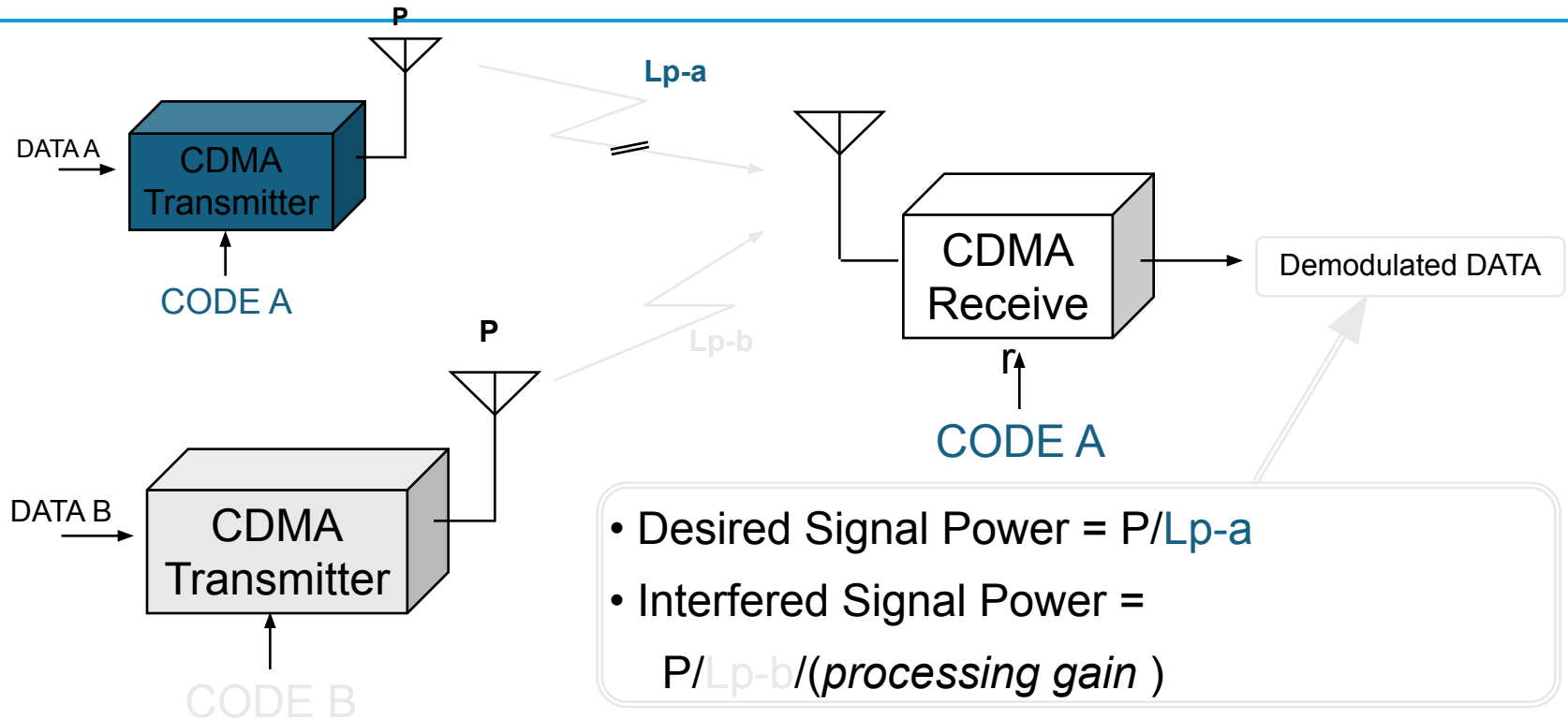
Fading in CDMA System (continued)

In CDMA system, multi-path propagation improves the signal quality by use of RAKE receiver.



Less fluctuation of detected power, because of adding all energy.

Near-Far Problem



When user B is close to the receiver and user A is far from the receiver, L_{p-a} could be much bigger than L_{p-b} . In this case, desired signal power is smaller than the interfered power.

Power Control on Forward Link and Reverse Link

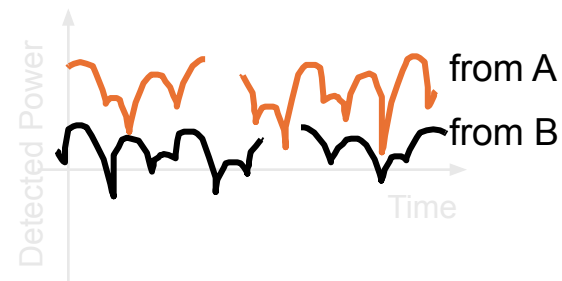
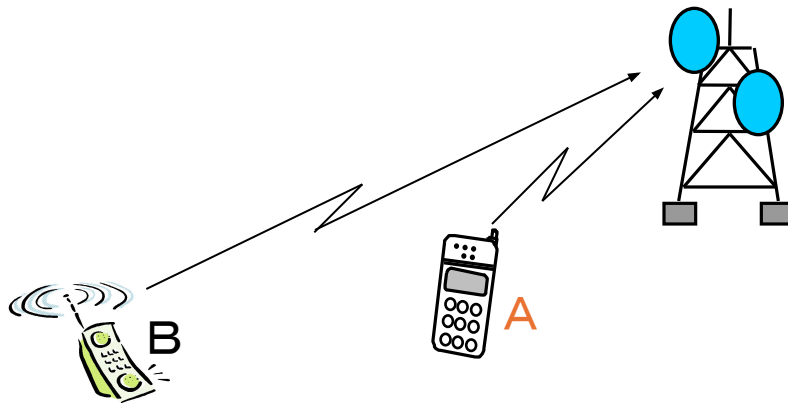
- On Forward Link
 - to send just enough power to reach users at the cell edge
- On Reverse Link
 - to overcome the 'near-far' problem in DS-CDMA

Power Control...

When all mobile stations transmit the signals at the same power (MS),
the received levels at the base station are different from each other,
which depend on the distances between BS and MSs.

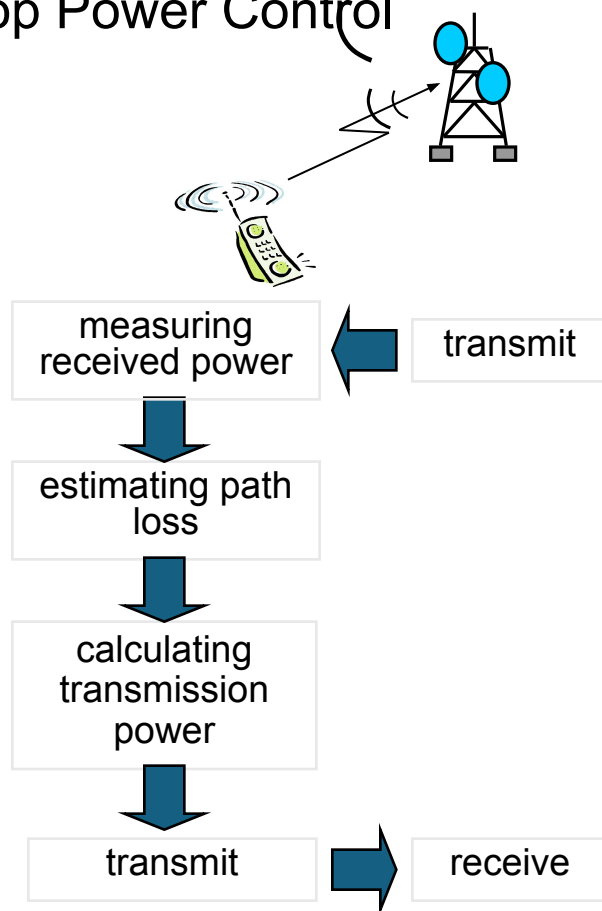
Moreover, the received level fluctuates quickly due to fading.

In order to maintain the received level at BS, power control technique must be employed in CDMA systems.

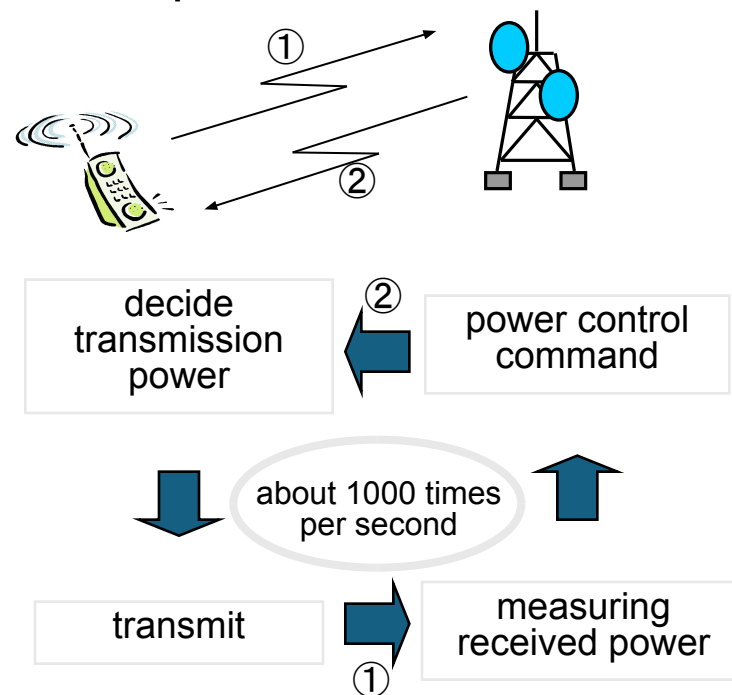


Power Control (continued)

Open Loop Power Control



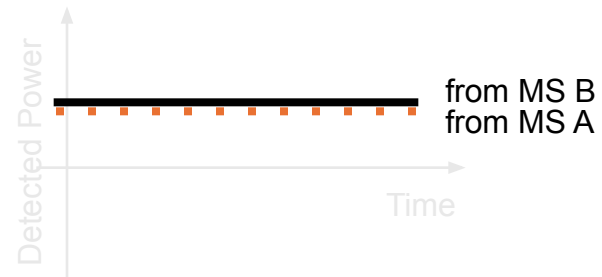
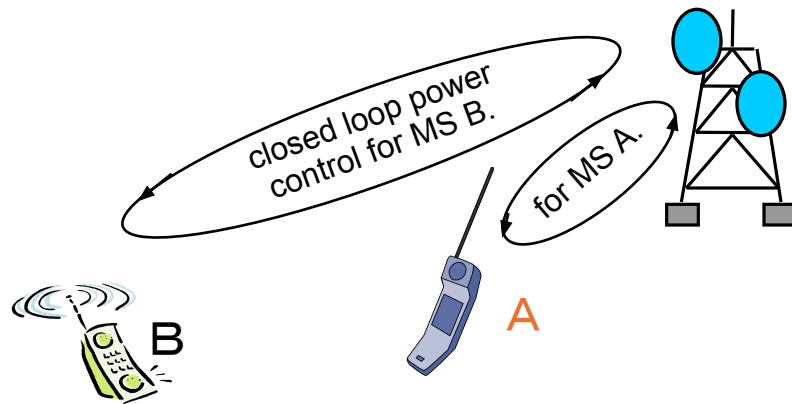
Closed Loop Power Control



Effect of Power Control

Effect of Power Control

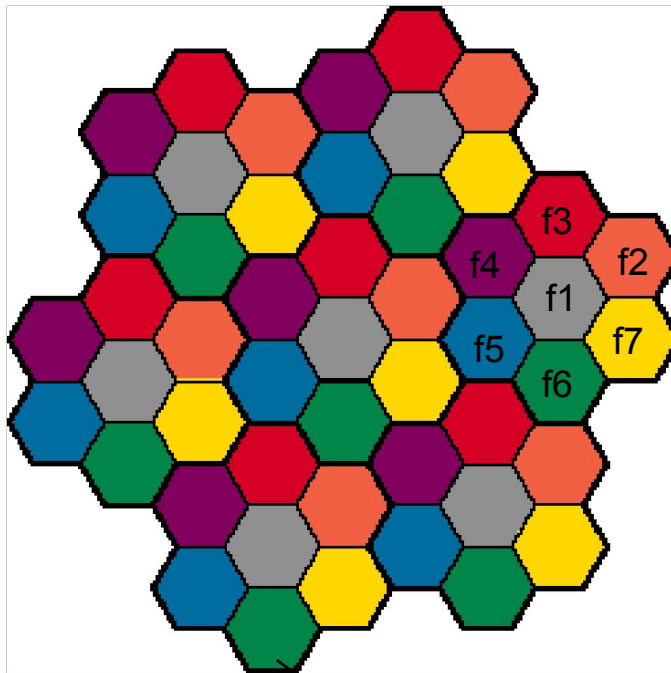
- Power control is capable of compensating the fading fluctuation.
 - Received power from all MS are controlled to be equal.
- ... Near-Far problem is mitigated by the power control.



Frequency Allocation (1/2)

In FDMA or TDMA,

radio resource is allocated not to interfere among neighbor cells.



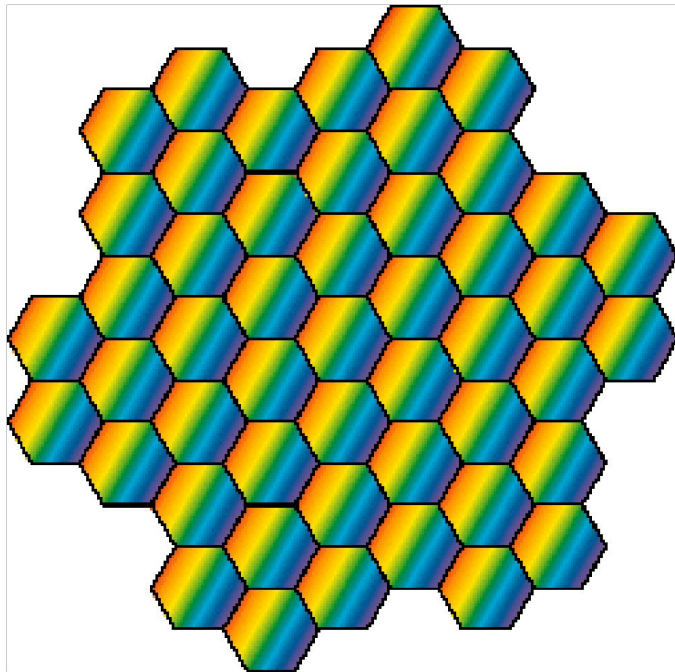
cell :
a "cell" means covered area by one base station.

- Neighbor cells cannot use the same (identical) frequency band (or time slot).
- The left figure shows the simple cell allocation with seven bands of frequency.
- In actual situation, because of complicated radio propagation and irregular cell allocation, it is not easy to allocate frequency (or time slot) appropriately.

Frequency Allocation (2/2)

In CDMA,

identical radio resource can be used among all cells,
because CDMA channels use same frequency simultaneously.



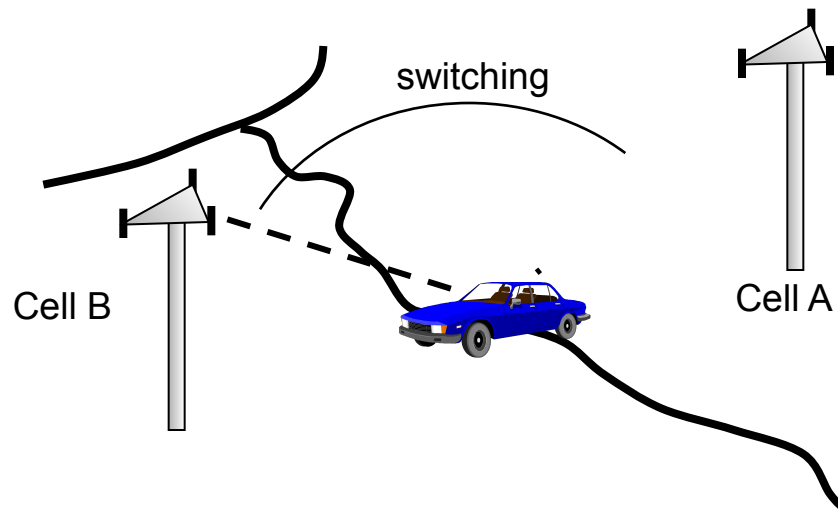
- Frequency allocation in CDMA is not necessary.
- In this sense, CDMA cellular system is easy to be designed.

Soft Handoff (1/2)

Handoff :

- Cellular system tracks mobile stations in order to maintain their communication links.
- When mobile station goes to neighbor cell, communication link switches from current cell to the neighbor cell.

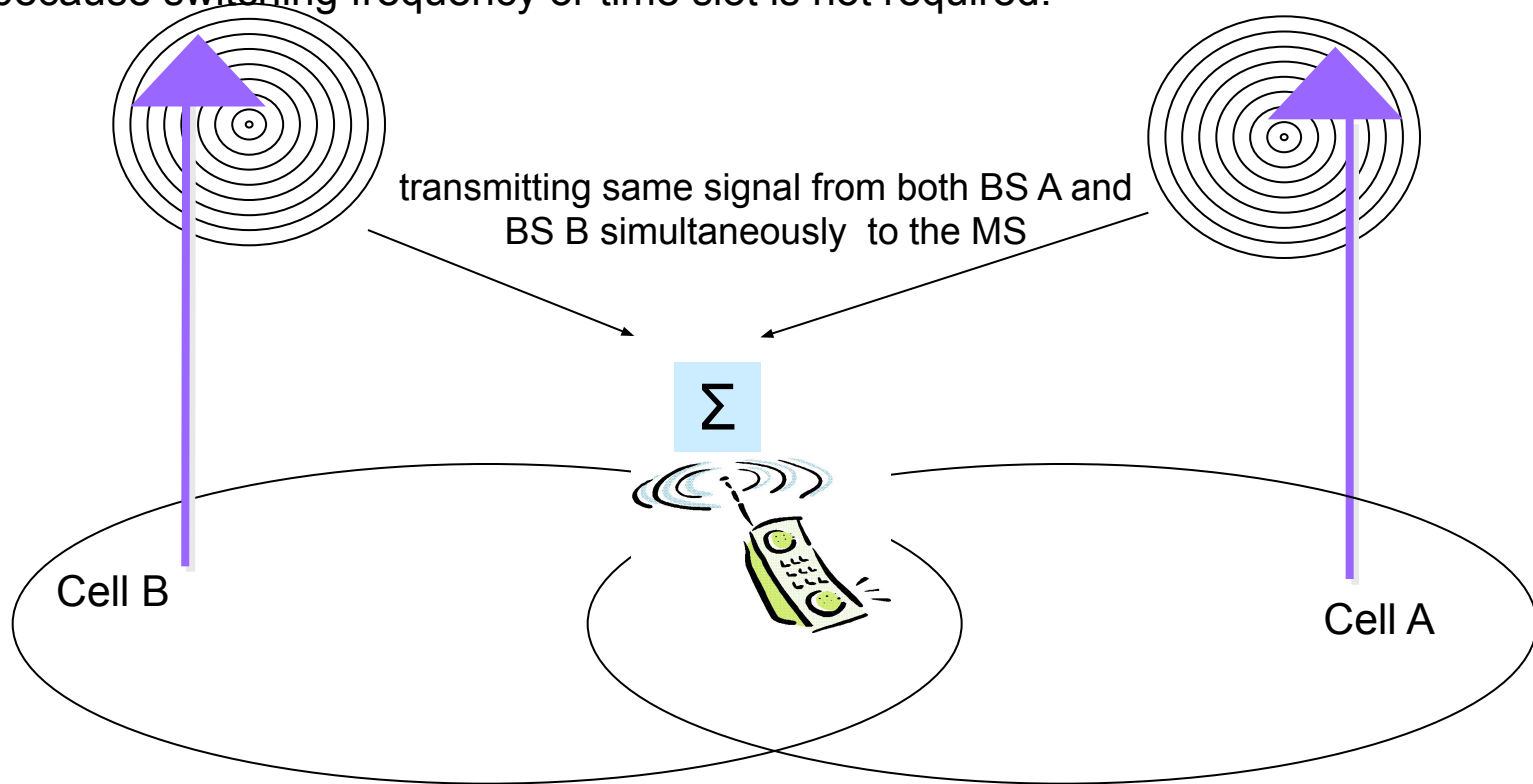
Hard Handoff : In FDMA or TDMA cellular system, new communication establishes after breaking current communication at the moment doing handoff. Communication between MS and BS breaks at the moment switching frequency or time slot.



Hard handoff : connect (new cell B) after break (old cell A)

Soft Handoff (2/2)

In CDMA cellular system, communication does not break even at the moment doing handoff, because switching frequency or time slot is not required.



Soft handoff : break (old cell A) after connect (new cell B)

Handoff in CDMA System

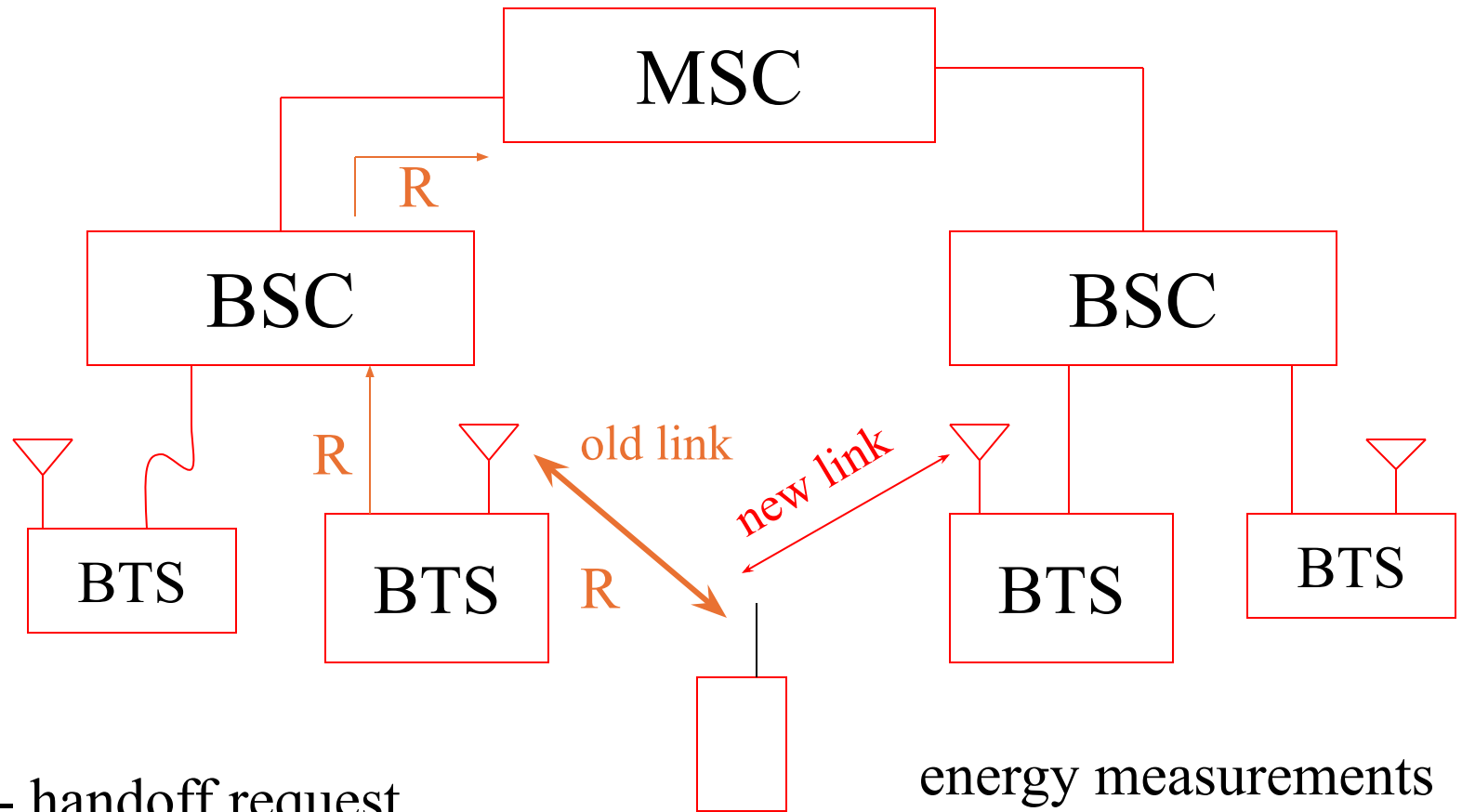
- Soft Handoff
 - Mobile commences Communication with a new BS without interrupting communication with old BS
 - same frequency assignment between old and new BS
 - provides different site selection diversity
- Softer Handoff
 - Handoff between sectors in a cell
- CDMA to CDMA hard handoff
 - Mobile transmits between two base stations with different frequency assignment

Soft Handoff- A unique feature of CDMA Mobile

Advantages

- Contact with new base station is made before the call is switched
- Diversity combining is used between multiple cell sites
 - additional resistance to fading
- If the new cell is loaded to capacity, handoff can still be performed for a small increase in BER
- Neither the mobile nor the base station is required to change frequency

Soft Handoff Architecture



R- handoff request
sent to the old cell

energy measurements
are made at the mobile

Conclusion

- CDMA is based on the spread spectrum technique which has been used at military field.
- CDMA cellular system is deemed superior to the FDMA and TDMA cellular systems for the time being.
- Therefore, CDMA technique becomes more important in radio communication systems.

Final Points on FDMA/TDMA/CDMA

- When users are in the middle of a phone call, the system uses FDMA/TDMA/CDMA to give them access.
- But there are only so many frequencies, time-slots, or codes available to share between users in a cell.
- If we divide the frequency into too many bands, or use too many time slots, or too many codes, the quality of speech heard by the end user will be unsatisfactory.

Orthogonal Frequency Division Multiplexing

- Definition
- Generation Of OFDM Signal
- Pros and cons
- OFDM and the physical layer of communication

OFDM

Orthogonal Frequency Division Multiplexing

- OFDM is a special case of FDM
- It's a Multi-carrier modulation scheme
(MultiRate, Multisymbol, Multicarrier)

Modulation – Mapping of the information by changing the carrier's phase, frequency of amplitude of combination

Multiplexing – Method of sharing a bandwidth amongst many independent channels.

- The tx bandwidth is divided into many narrow subchannels which are transmitted in parallel.
- Each subchannel is narrow enough so that the fading it experiences is flat ☐ **no ISI**.

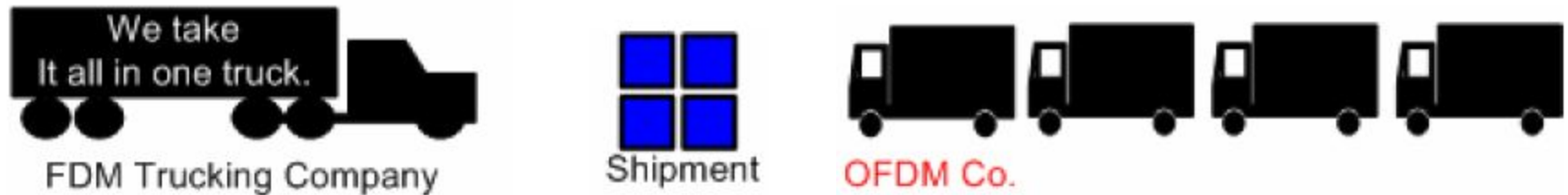


FDM



OFDM

Analogy Example---- Mapping & Multiplexing

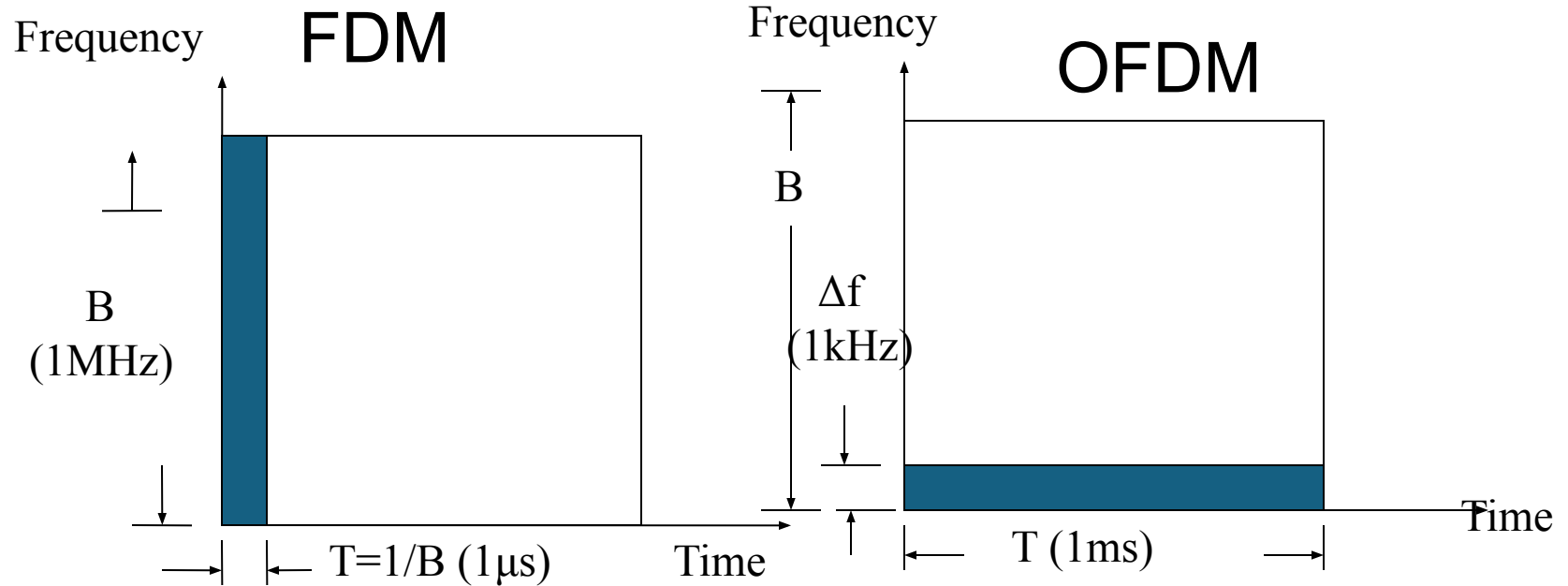


- Both methods will carry exactly same amount of data , But in case of accident ,only $\frac{1}{4}$ of data on the OFDM truck will suffer.
- These four trucks when we see as a signal are called **sub-carriers** in OFDM system and they must be orthogonal to each other for this system to work efficiently.
- The independent truck/multi-carriers can be multiplexed by frequency division multiplexing (FDM) , called multi-carrier transmission or can be based on **code division multiplexing (CDM)**, called **multi-code transmission**.



Basic Difference

In FDM the question of multiplexing is applied to independent signals but these independent signals are a sub-set of the one main signal. In OFDM the signal itself is first split into independent channels, modulated by data and then re-multiplexed to create the OFDM carrier.

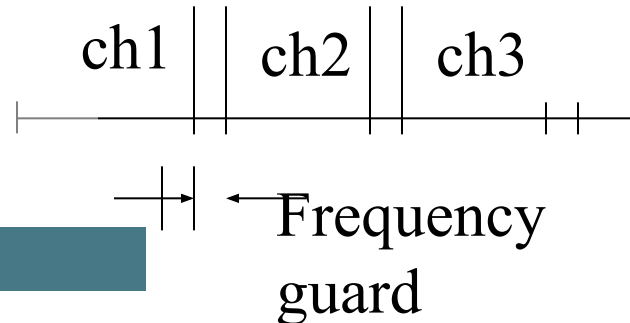


Single Carrier System

Bandwidth = B
Symbol Duration $T = 1/B$

When $\tau_{\max} > T \rightarrow$ Intersymbol interference (ISI)

Equaliser is needed to reduce ISI



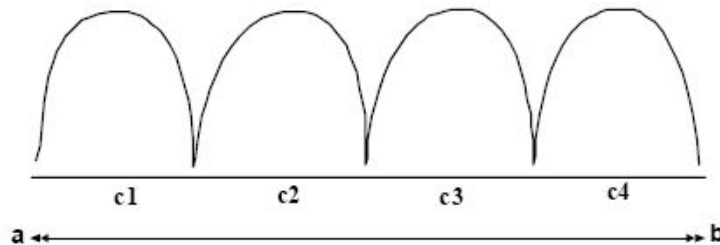
- Available bandwidth is divided into N subband
- Each symbol occupies a narrow band but longer time period

For each sub carriers Bandwidth $\Delta f = B/N$ Symbol duration $T = N/B$

OFDM a Special case Of FDM

FDM:-

- Whole of the bandwidth in FDM is sub-divided into space of n equal spaces



- Frequencies a & b may or may not be integer because there is no relationship implied between a & b.
- Same is true for center frequencies of each carrier as no relationship implied at center frequencies

OFDM:-

- Now suppose that the above carriers C_1 and C_n are such that, C_n is the integral multiple of C_1 .

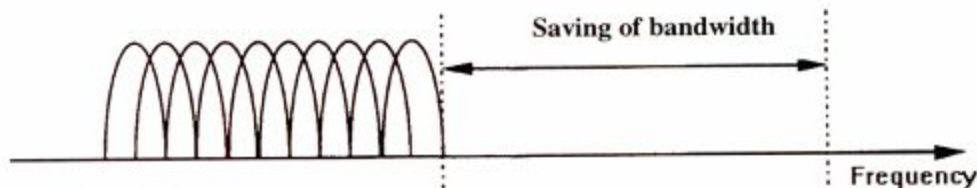
$$\text{i.e. } C_n = n * C_1,$$

then both the frequencies are ORTHOGONAL to each other.

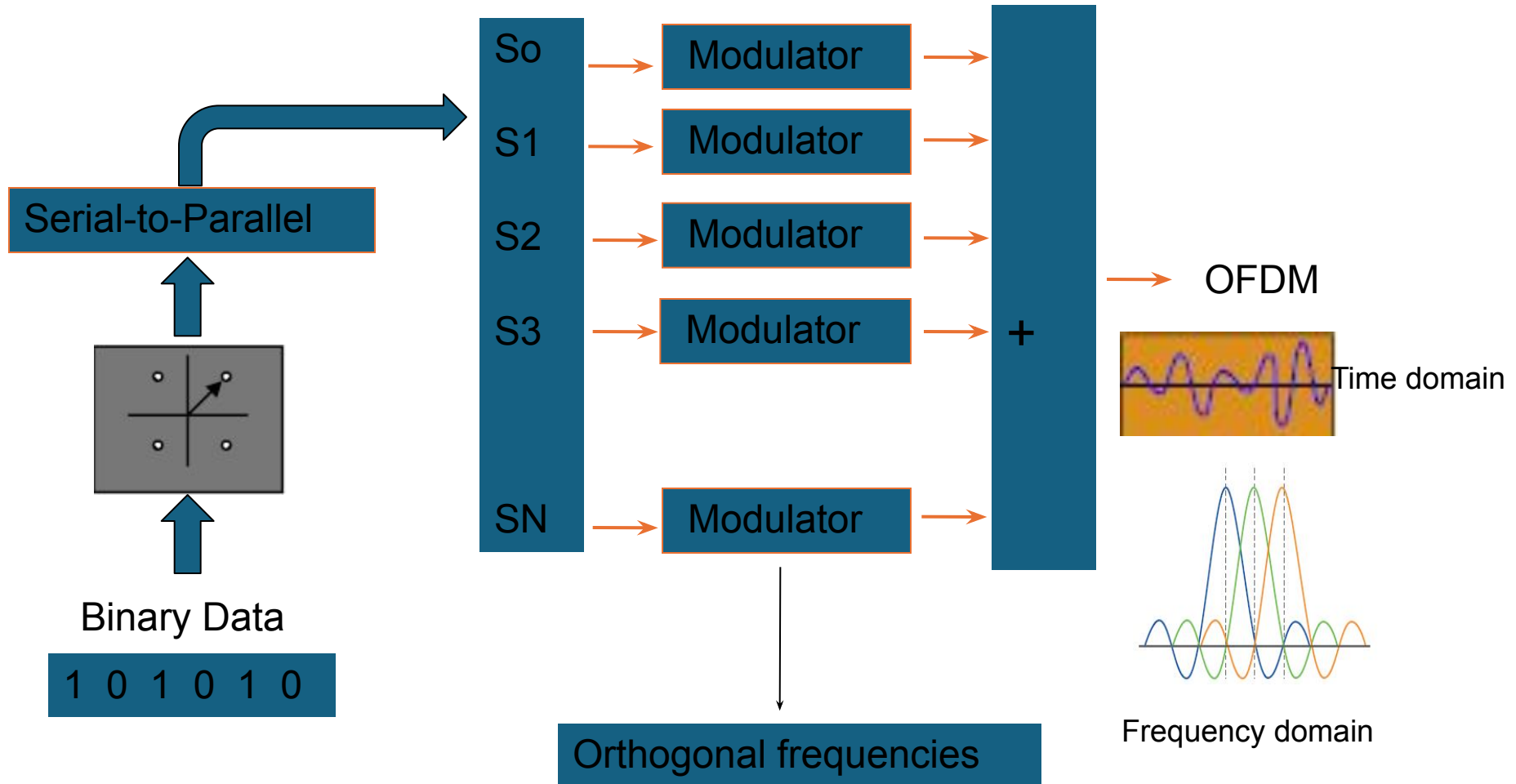
- Hence

- $C_2 = 2 C_1$
- $C_3 = 3 C_1$
- $C_4 = 4 C_1$

All these frequencies C_2, C_3 and C_4 are called the **harmonics** of C_1 and are ORTHOGONAL to each other.



How to Generate and OFDM signal?



Example:

- Suppose we transmit a signal having 4 sub-carriers to see the development of OFDM signal.
- The signal has a symbol rate of 1 and sampling frequency of 1 symbol per sample, so each transmission is a bit.
- Suppose the bit stream to be transmitted is:

1, 1, -1, -1, 1, 1, 1, -1, 1, -1, -1, -1, -1, 1, -1, -1, -1, 1,...

- First step is to convert this serial data to parallel four columns.

– Serial to parallel conversion of data bits.

c1	c2	c3	c4
1	1	-1	-1
1	1	1	-1
1	-1	-1	-1
-1	1	-1	-1
-1	1	1	-1
-1	-1	1	1

- Each column is carried by one sub-carrier

What should be the Frequency of each carrier to carry respective data streams??

- According to Niquist Sampling Theorem:-

- “The smallest frequency that can carry the information has to be twice the information rate”

- Here the information rate per carrier is $\frac{1}{4}$.

- Therefore

- Min Frequency required = $\frac{1}{4} * 2 = \frac{1}{2}$ Hz

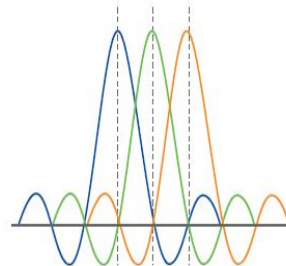
Lets take 1 Hz

- The harmonics of 1 Hz are:

- $1 * 2 = 2$ Hz
- $1 * 3 = 3$ Hz
- $1 * 4 = 4$ Hz

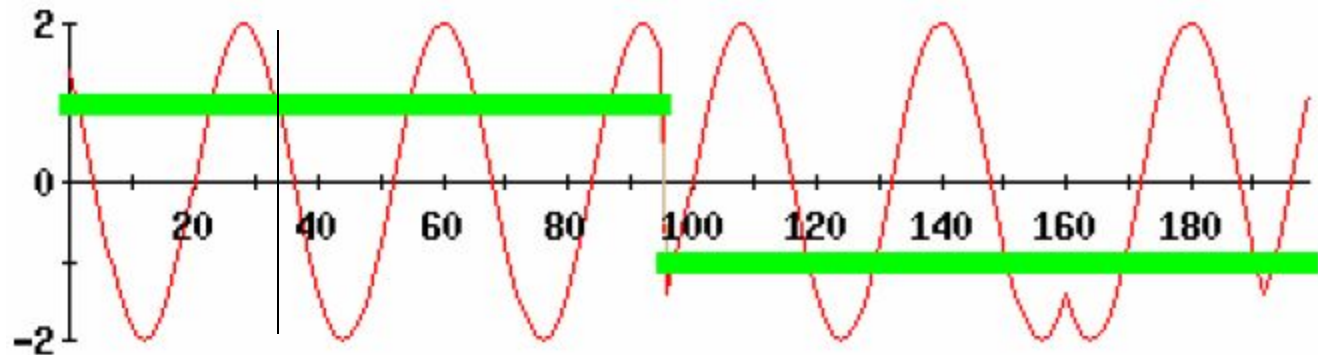
Had I picked $\frac{1}{2}$ Hz as my starting frequency then harmonics would be

- $\frac{1}{2} * 2 = 1$ Hz
- $\frac{1}{2} * 3 = \frac{3}{2}$ Hz
- $\frac{1}{2} * 4 = 2$ Hz

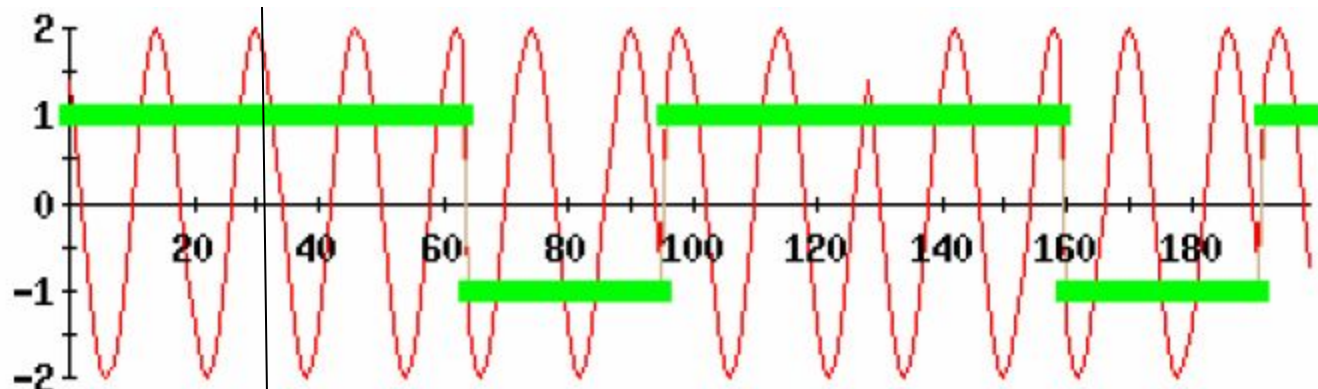


Modulation OR Mapping

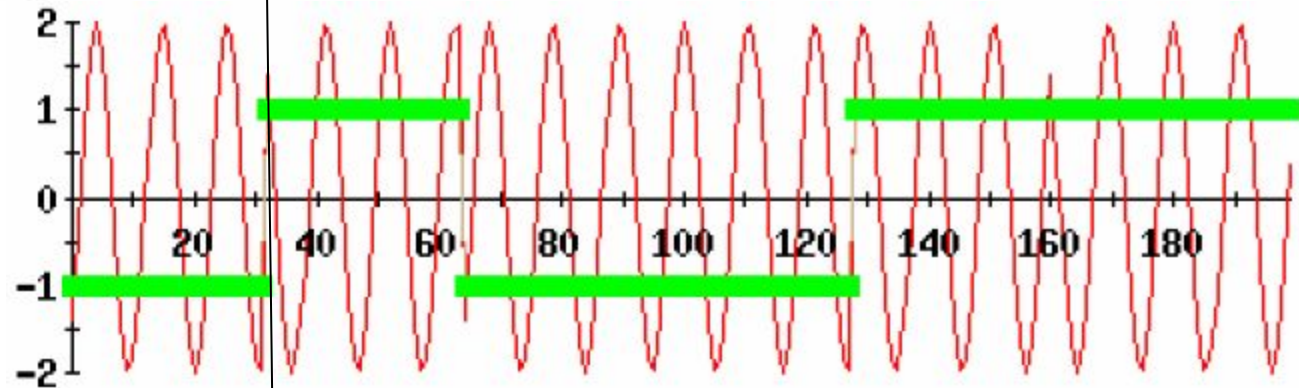
- Let us suppose BPSK as our modulation scheme for this example.
- So carrier 1 (1 1 1 -1 -1 -1) is to be mapped onto BPSK carrier frequency of 1 Hz.



Carrier 2



Carrier 3

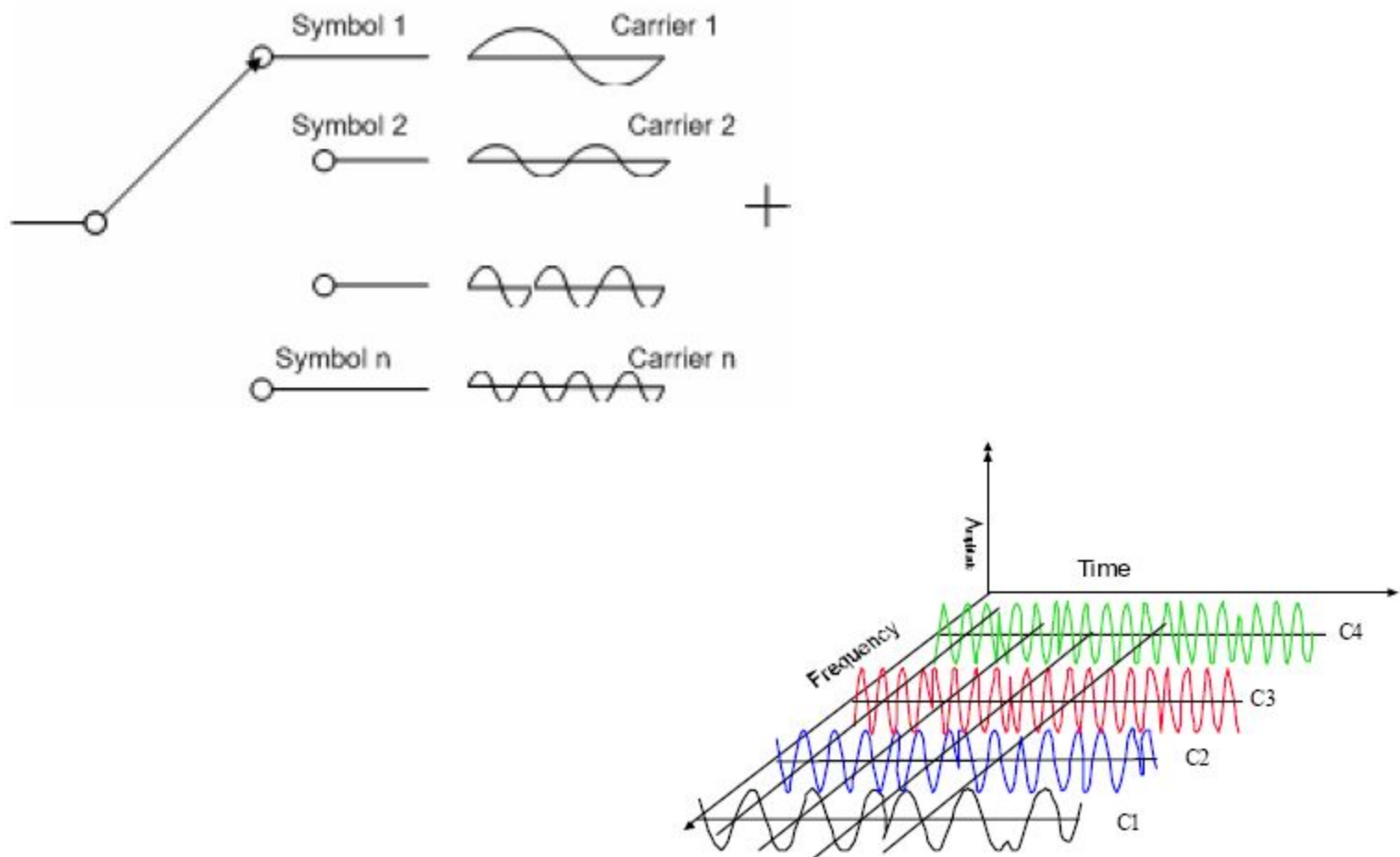


Carrier 4

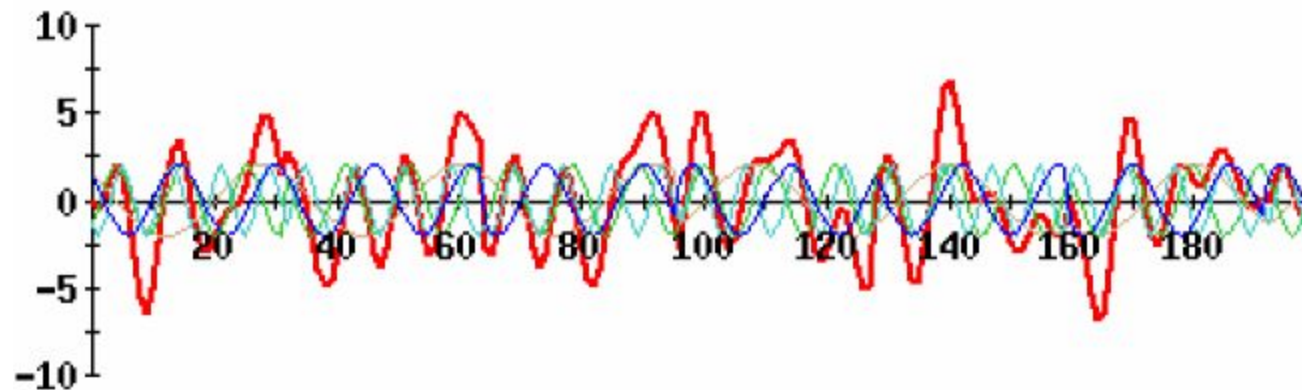


All the bits have been modulated on the orthogonal frequencies from 1 – 4 Hz.

- What basically we have done is, distributed the bits one at a time to four sub-carriers.



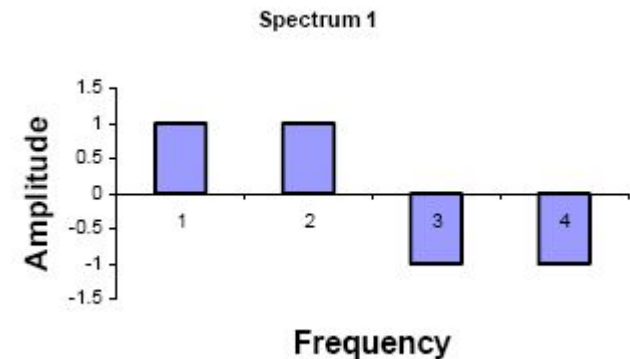
- Now Add all four modulated carriers to create the OFDM signal, often produced by block called IFFT.



IFFT/FFT

- IFFT is a mathematical concept. It crunches the in coming signal into a correct time domain result.
- Each row spectrum of our example has 4 frequencies and each of these rows can be converted to produce a time domain signal.

- IFFT quickly computes the time domain signal instead of having it to do one carrier at a time and then adding.



The functional block diagram of how the signal is modulated/demodulated is given below.



IFFT & FFT Equations.

D_m represent the time domain composite signal containing all the orthogonal frequencies after modulation.

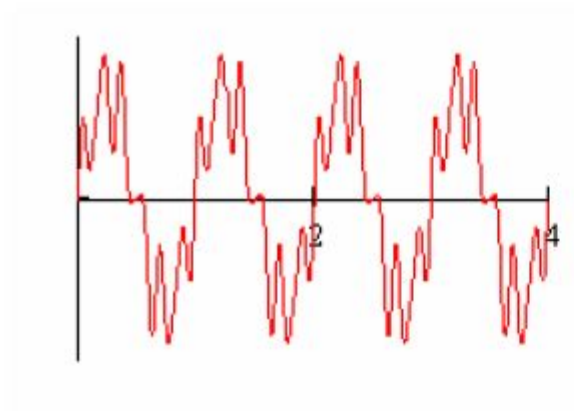
S_n are the complex constellation symbols. The m denotes the time domain sample number.

N refers the number of orthogonal harmonics

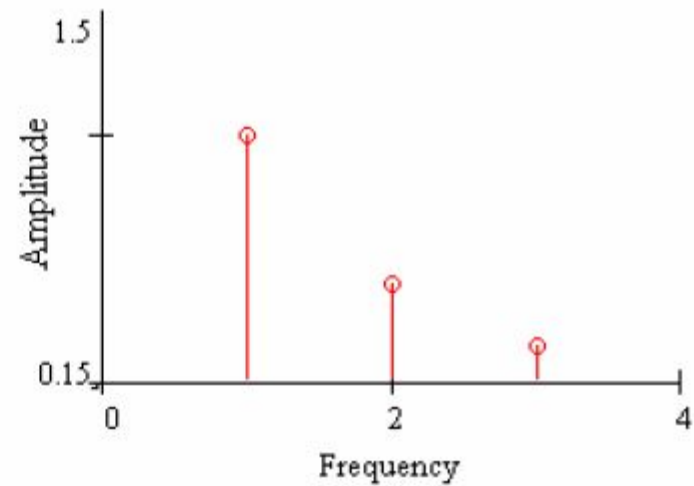
$$D_m = \text{IFFT} \{S_n\} = \frac{1}{N} \sum_{n=0}^{N-1} S_n e^{j(2\pi/N) nm}$$

$$S_n = \text{FFT} \{D_m\} = \sum_{m=0}^{N-1} D_m e^{-j(2\pi/N) nm}$$

FFT/IFFT



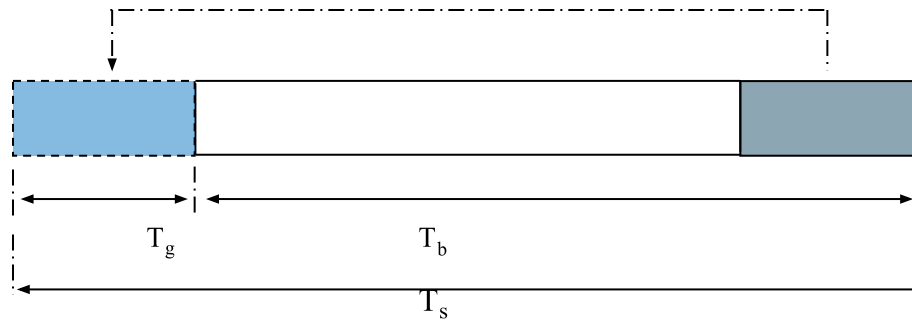
(a) Time domain view



(b) Frequency domain view

Cyclic Prefix

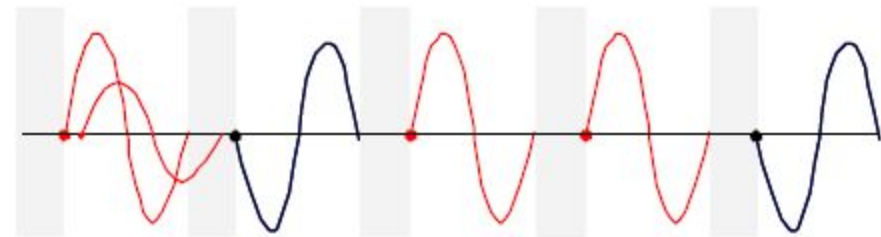
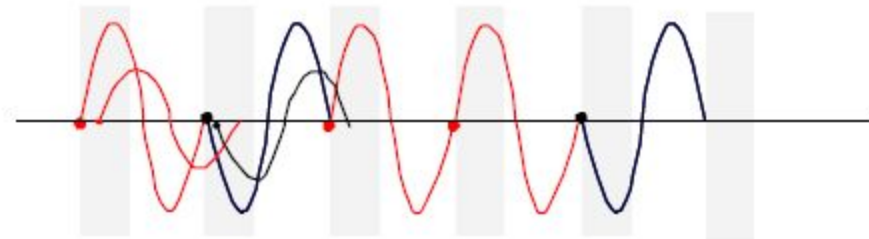
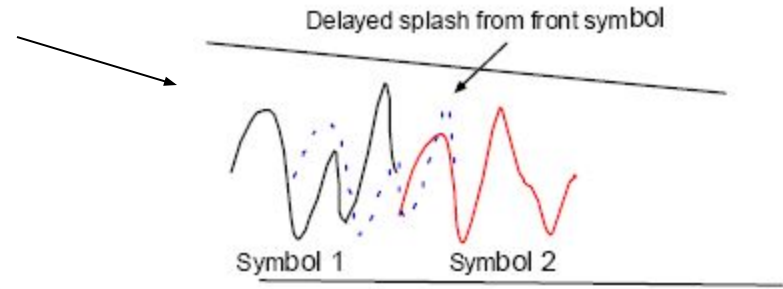
It is a copy of the last portion of the time domain OFDM waveform which is transmitted first before the effective part of the OFDM symbol



- Cyclic prefix is used in OFDM to reduce inter-symbol interference (ISI)
- Cyclic prefix is used in OFDM to reduce inter-carrier interference (ICI) introduced by the time dispersive multi-path channel
- It preserves the orthogonality of sub-carriers.

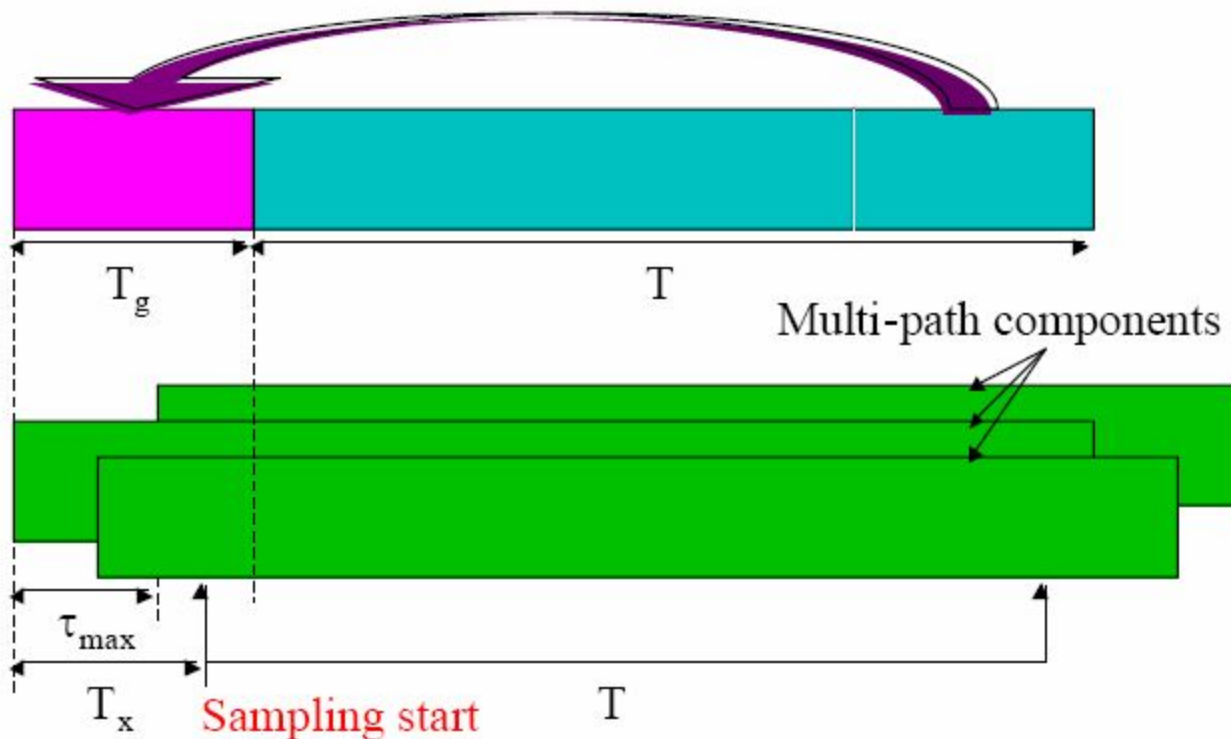
Cyclic Prefix

- The first symbol and its spreaded version
- The PSK symbol and its delayed Version.
- This delayed version becomes noise for the next symbol and destroys the beginning bits of the next symbol.
- To mitigate the noise in the beginning of the next symbol, we will move the symbol further away from the region of delay spread

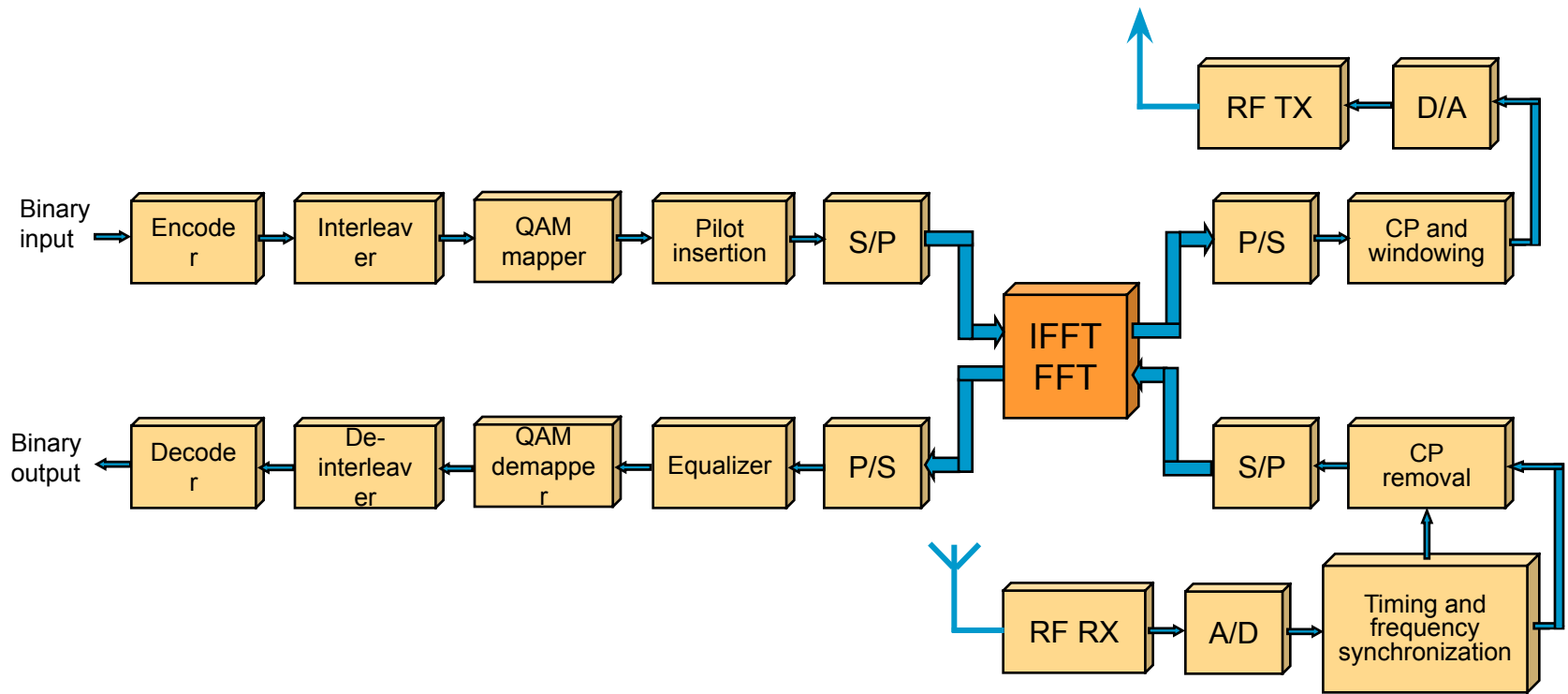


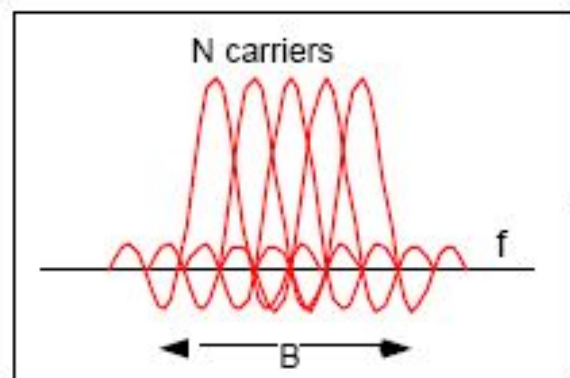
Introducing Cyclic Prefix

- So to introduce cyclic prefix, slide the symbol away to start at the edge of delay spread time and then
- Fill the guard space with the copy of the tail of this symbol

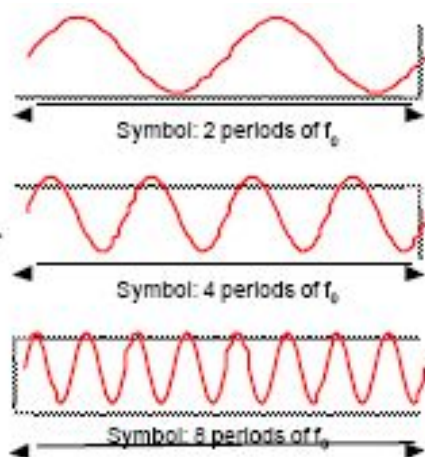


OFDM Transceiver



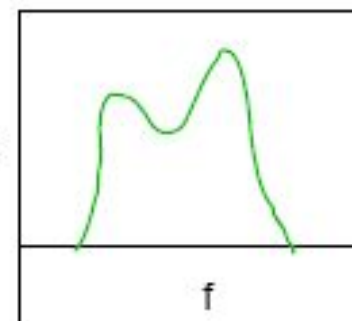


Data coded in frequency domain



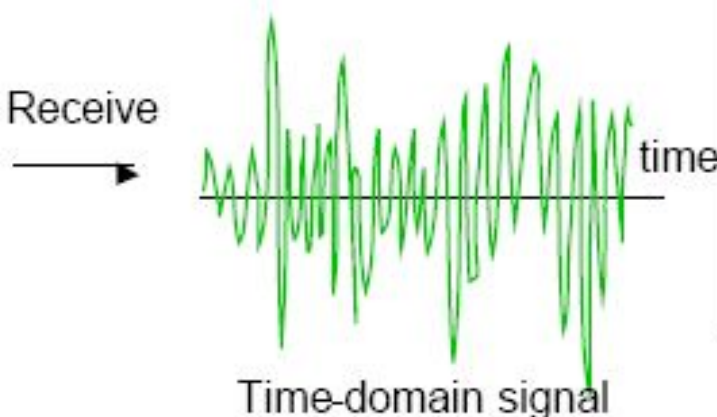
Transformation to time domain:
each frequency is a sine wave
in time, all added up.

Transmit

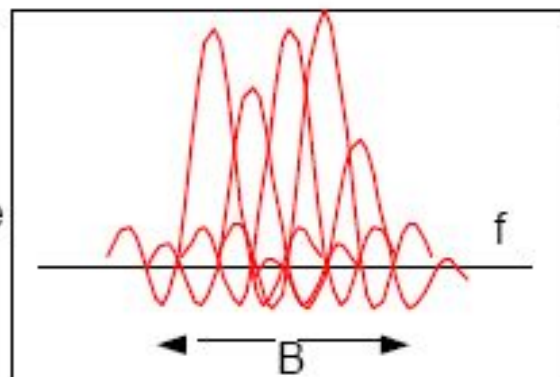


Channel frequency
response

Receive



Time-domain signal



Frequency-domain signal

Decode each frequency
bin separately

Applications

- High-speed digital subscriber lines (HDSL, ADSL, VDSL)
- Digital broadcasting systems (DAB, DVB-T)
- Wireless LANs (HIPERLAN/2, IEEE 802.11a, IEEE 802.16, MMAC)
- Fixed broadband wireless access (IEEE 802.11g)
- Considered for wireless ATM network (Magic-WAND),
- Considered for future broadband radio area networks and multimedia communications (European and American projects).

OFDM Advantages

- Robustness against multipath fading and intersymbol interference and narrow-band interference .
- High spectral efficiency, no side bandwidth required for system operation,
- Simple frequency domain equalization
- It handles the frequency selective channels well.

OFDM Disadvantages

- High Peak-to-Average Power Ratio (PAPR) of the transmitted signal resulting in:
 - clipping noise (limited quantization levels, rounding and truncation, during IFFT and FFT computation)
 - nonlinear distortions of power amplifiers
 - BER performance degradation,
 - energy spilling into adjacent channels,
- Peak-to-Average problem reduces the power efficiency of RF amplifier at the transmitter
- Sensitivity to frequency offset and phase noise, which are sources of inter carrier interference,